

MapleCFO: an AI personal-finance agent for Canadians

EECS3311 Software Design, Fall 2026: Course Project, Stage 1 (Design Report)

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Project	MapleCFO, an AI "Chief Financial Officer" agent
Language / platform	Java 21 · JavaFX (GUI) · picocli (CLI) · SQLite · JUnit 5
AI model(s)	Anthropic Claude (Messages API with tool use); a scripted <code>MockLlmClient</code> for offline demos and tests
Stage	1: Design (before implementation)

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1. Project overview

1.1 Problem and motivation

Canadian students and young professionals manage their money across several banks, credit cards and **registered accounts (TFSA, RRSP, FHSA)**. Every account has its own app, and the questions that actually matter cut across all of them:

- "Can I afford a \$2,000 trip in December without falling behind on my laptop fund?"
- "Should my next \$500 go into my TFSA or toward my credit card?"
- "How much TFSA room do I actually have? I don't want an over-contribution penalty."
- "When could I retire if I saved 25% instead of 15%?"

Banking apps only show one institution. Spreadsheets need manual formulas. Generic chatbots **invent numbers** because they cannot see the user's data. No tool combines the user's real data, reliable Canadian calculations and plain-language explanations.

1.2 Target users

User group	Typical needs
University and college students	Budgeting on a small income, subscriptions, first credit card, first TFSA
Young professionals (22–35)	Net worth, TFSA/RRSP/FHSA room, saving for a first home, debt payoff, FIRE planning
Newcomers to Canadian personal finance	Understanding how registered accounts, budgeting and debt payoff work

1.3 What the agent does

MapleCFO is a desktop application with a **GUI and a CLI** that imports the user's bank and card CSV exports, organises them, and runs a set of deterministic financial calculators. On top of this sits the **AI CFO agent**. It:

1. **Interprets** a free-form money question.
2. **Plans** which information it needs (spending, contribution room, goals, cash-flow forecast...).
3. **Uses tools**: it calls deterministic finance tools (10 of them) through a tool registry, one or more times.
4. **Reasons** over the tool results and explains the trade-offs in plain language, quoting only numbers the tools returned.
5. **Remembers** the conversation, so follow-ups like "what if I wait until January?" work.
6. **Proposes actions** (e.g. "create a savings goal") that the user must **confirm** before anything changes.

1.4 Why an AI agent is appropriate

- **The questions are open-ended and multi-step.** "Can I afford X?" needs a cash-flow forecast, the committed savings goals and budget status. A fixed form cannot anticipate every combination; an agent that plans and calls tools can.
- **Messy data needs judgement.** Merchant strings like `SQ *DOLLARAMA #1234 TORONTO` are hard to categorise with rules alone. The LLM handles the long tail.
- **People need explanations, not just numbers.** The LLM turns calculator output into an explanation a non-expert can act on.
- **The design keeps the AI honest.** Every number comes from deterministic, unit-tested code. The LLM never does arithmetic, and a `ResponseValidator` checks that the dollar amounts in its answer appear in the tool results.

1.5 AI / LLM model(s)

Role	Model	Where used
Primary reasoning and tool-use model	Anthropic Claude (Sonnet-class model through the Messages API with tool use; exact version fixed at implementation time and recorded in Stage 2)	CfoAgent, LlmCategorizer, ReportSummarizer
Offline / test double	MockLlmClient (scripted, deterministic responses)	Unit and integration tests; demo mode when no API key is configured

Both sit behind the `LlmClient` interface (Adapter pattern), so the rest of the system does not know which one is in use, and another provider could be added later without changing the agent.

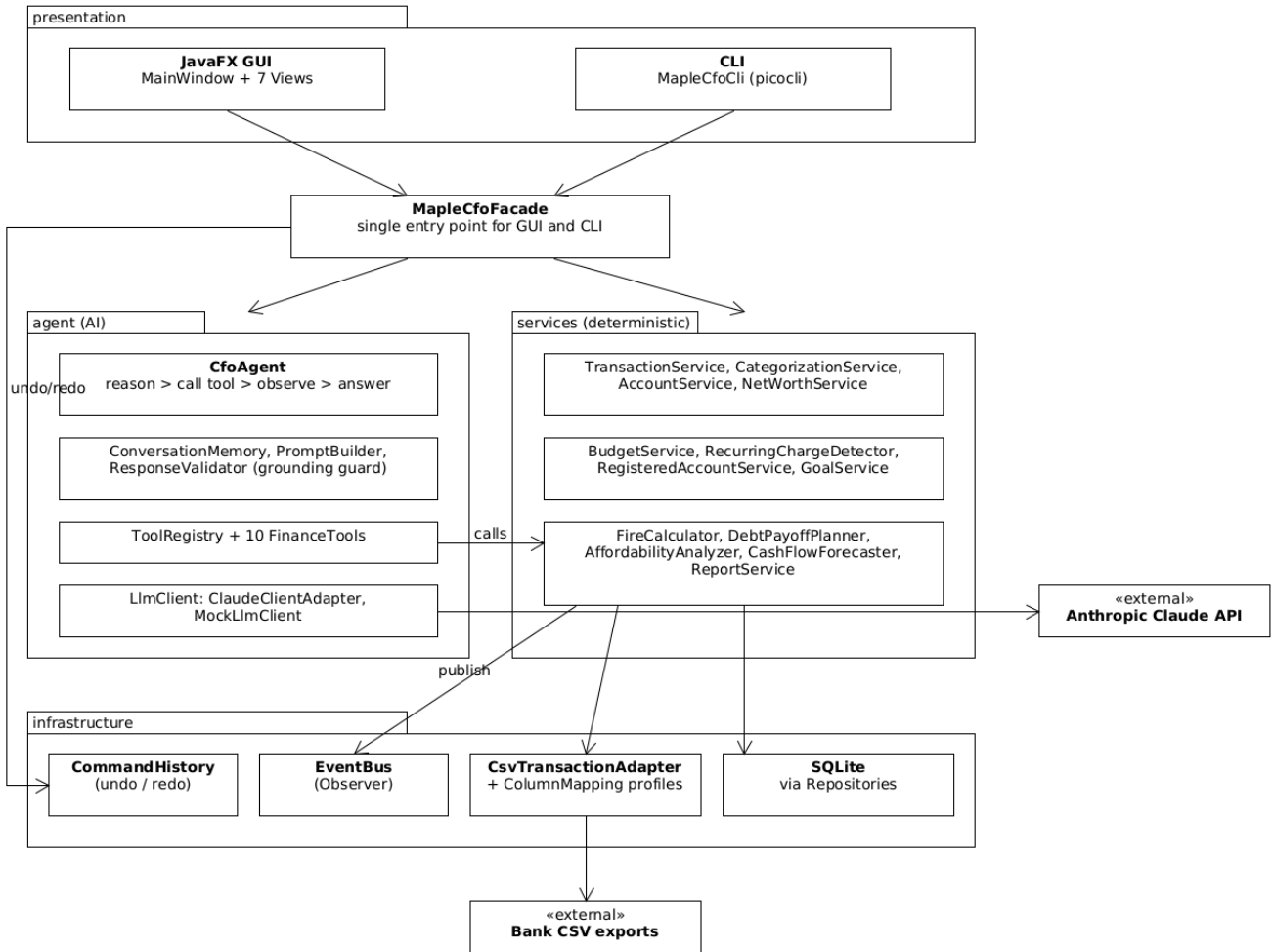
1.6 How the AI interacts with the rest of the system

- The LLM **never touches the database or the file system directly**. It can only request one of the registered `FinanceTools` by name, with JSON arguments.
- `ToolRegistry` **validates the arguments** against the tool's schema, runs the tool (which calls a deterministic service), and returns a structured `ToolResult` (or an error) to the agent.
- Tools that would **change data** (create a goal, set a budget) only create a *pending* `UndoableCommand`. It is executed only when the user clicks **Confirm**, and it can then be undone.
- The LLM is also used in two narrow, bounded places. It categorises transactions that no rule matched (and only answers that are valid category names are accepted), and it writes the narrative part of the monthly report, using figures that were already computed.

1.7 Overall architecture

The system is organised in layers. The GUI and CLI are thin presentation layers that talk **only** to `MapleCfoFacade`. The facade coordinates the deterministic domain services, the undo/redo command history and the AI agent. Infrastructure (SQLite repositories, the CSV import adapter, the event bus and the LLM adapter) sits at the bottom.

Architecture: layered package view



Layer	Responsibility	Main classes
Presentation	Show data, collect input (GUI and CLI)	MainWindow, 7 Views, MapleCfoCli, CliRenderer
Application	Single entry point, undo/redo, event distribution	MapleCfoFacade, CommandHistory, EventBus
Agent (AI)	Plan, call tools, remember, validate answers	CfoAgent, ToolRegistry, FinanceTools, ConversationMemory, PromptBuilder, ResponseValidator, LlmClient
Domain services (deterministic)	All financial logic and calculations	TransactionService, BudgetService, RegisteredAccountService, FireCalculator, DebtPayoffPlanner, ...
Infrastructure	Persistence, import, external APIs	Repository<T, ID> + SQLite implementations, CsvTransactionAdapter, ClaudeClientAdapter

2. Detailed feature specifications

MapleCFO has **12 major features**. The scope is deliberately kept lean so that every feature can be fully built and tested within the course timeline. Login, settings and "about" screens are not counted. Each feature is classified as **Deterministic** (no AI), **AI** (driven by the LLM) or **Hybrid** (deterministic core, AI for one bounded step).

ID	Feature	Type
F01	Bank/card CSV import with column-mapping profiles	Deterministic
F02	Smart transaction categorization with correction and undo	Hybrid
F03	Accounts and net-worth dashboard	Deterministic
F04	Monthly budgets and overspending alerts	Deterministic
F05	Recurring charge and subscription detection	Deterministic
F06	TFSA / RRSP / FHSA contribution-room tracker	Deterministic
F07	FIRE (financial independence) projection	Deterministic
F08	Debt payoff planner (avalanche vs snowball)	Deterministic
F09	Savings goals with forecasts	Deterministic
F10	AI CFO chat (multi-step, tool-using agent with memory)	AI
F11	"Can I afford it?" affordability check	Hybrid
F12	Monthly summary with AI narrative	Hybrid

F01: Bank/card CSV import with column-mapping profiles

Field	Specification
Description	Imports transactions from any bank or card CSV export through one generic importer . The first time, the user maps the file's columns (date, description, amount or debit/credit) and saves the mapping as a named profile (e.g. "TD chequing"). Later imports reuse the profile. Invalid rows are rejected with reasons, and duplicates from overlapping exports are skipped.
User interaction (GUI)	<i>Transactions</i> tab → Import CSV → file chooser, account selector, profile drop-down → column-mapping dialog when needed → import summary dialog. CLI: <code>maplecfo import <file> --account <id> --profile "TD chequing"</code> .
Input	CSV file path; target account; mapping profile name (or a new mapping).
Output	<code>ImportResult</code> (number imported, duplicates skipped, rejected rows with reasons, number left uncategorized) and the new rows in the transaction table.
AI involvement	Deterministic (categorization afterwards is F02).
Expected workflow	Load the profile → check that it fits the file's header → <code>CsvTransactionAdapter</code> reads, parses and validates rows (date, amount, description) → deduplicate using a fingerprint (date + amount + normalised description + account) → categorize (F02) → save → publish <code>TRANSACTIONS_IMPORTED</code> .
Error / alternative cases	No profile, or the profile does not match the header → the mapping dialog opens and the new mapping can be saved. Unreadable or empty file → error message, nothing saved. Some bad rows → valid rows are imported and bad rows are listed. File already imported → every row is reported as a duplicate.

F02: Smart transaction categorization with correction and undo

Field	Specification
Description	Assigns every transaction a category (Groceries, Dining, Rent, ...). Rules run first (merchant keywords and rules learned from the user's past corrections). Only transactions no rule matches are sent to the LLM, which must answer with one of the allowed category names. The user can correct any category, the system learns a rule from the correction, and every correction can be undone or redone .
User interaction (GUI)	Category drop-down in each row of the <i>Transactions</i> table; Undo/Redo buttons (Ctrl+Z / Ctrl+Y); filter "Needs review". CLI: <code>maplecfo tx recategorize <id> <CATEGORY></code> , <code>maplecfo undo</code> .
Input	Transactions (from F01), or a transaction id plus a new category.
Output	Categorized transactions with their source (RULE / AI / USER); updated budgets and dashboard.
AI involvement	Hybrid : rules first, LLM only as a fallback.
Expected workflow	<code>CategorizationService</code> tries the strategies in order: <code>RuleBasedCategorizer</code> , then <code>LlmCategorizer</code> . An AI answer is accepted only if it is exactly one of the <code>Category</code> names. Anything else leaves the transaction <code>UNCATEGORIZED</code> ("needs review"). A user correction runs as a <code>RecategorizeCommand</code> through <code>CommandHistory</code> and adds a merchant rule.
Error / alternative cases	LLM unavailable or times out → rule-only mode, the rest marked for review. LLM returns an unknown category or extra text → rejected. Transaction deleted before undo → undo reports failure and the stacks stay consistent.

F03: Accounts and net-worth dashboard

Field	Specification
Description	The user registers accounts (chequing, savings, credit cards, TFSA, RRSP, FHSA, loans) and keeps their balances up to date. The dashboard shows total assets, total liabilities and net worth, grouped (Banking, Registered, Investments, Debts), plus a net-worth history chart.
User interaction (GUI)	<i>Dashboard</i> tab: net-worth card, breakdown chart, Add account , and inline balance editing. CLI: <code>maplecfo accounts add ...</code> , <code>maplecfo networkth</code> .
Input	Account name, type, institution, balance.
Output	<code>NetWorthSnapshot</code> (total, assets, liabilities, subtotal per group) and charts.
AI involvement	Deterministic.
Expected workflow	Accounts are arranged as a Composite tree (<code>AccountGroup</code> → <code>Account</code>). <code>NetWorthService</code> asks the root for <code>getValue()</code> , which adds up its children recursively. Liabilities count as negative. A snapshot is stored each time balances change, for the history chart.
Error / alternative cases	Negative balance on an asset account → validation error. No accounts yet → empty state with an "Add your first account" prompt. Duplicate account name → warning.

F04: Monthly budgets and overspending alerts

Field	Specification
Description	The user sets a monthly limit per category. The system tracks spending against each limit and raises an alert when a category reaches 80% and 100% of its limit.
User interaction (GUI)	<i>Budget</i> tab: list of categories with progress bars, Set limit field, alert banner. CLI: <code>maplecfo budget set DINING 250</code> , <code>maplecfo budget status</code> .
Input	Category, monthly limit, month.
Output	<code>List<BudgetStatus></code> (spent, remaining, % used, exceeded?) and <code>BUDGET_EXCEEDED</code> alerts.

Field	Specification
AI involvement	Deterministic.
Expected workflow	Setting a budget runs a <code>SetBudgetCommand</code> (undoable). <code>BudgetService</code> subscribes to the <code>TRANSACTIONS_IMPORTED</code> and <code>TRANSACTION_RECATEGORIZED</code> events. After each event it recomputes the status and publishes <code>BUDGET_EXCEEDED</code> , which the GUI views and <code>AlertService</code> receive.
Error / alternative cases	Limit ≤ 0 or non-numeric $\rightarrow$ validation error. A budget already exists for that category and month $\rightarrow$ replaced (and the old one is restored on undo). No transactions yet $\rightarrow$ 0% used.

F05: Recurring charge and subscription detection

Field	Specification
Description	Finds charges that repeat at a regular interval (streaming, phone, gym, rent), estimates their yearly cost, predicts the next charge date, and flags price increases .
User interaction (GUI)	<i>Dashboard</i> $\rightarrow$ <i>Subscriptions</i> panel with a yearly total and "price increased" badges. CLI: <code>maplecfo recurring</code> .
Input	Transactions from the last 6 months.
Output	<code>List<RecurringCharge></code> (merchant, average amount, frequency, next expected date, <code>priceIncreased</code>).
AI involvement	Deterministic.
Expected workflow	Normalise merchant names $\rightarrow$ group by merchant $\rightarrow$ keep groups with at least 3 charges at a roughly regular interval (weekly, monthly or yearly, ± 3 days) $\rightarrow$ compute the average $\rightarrow$ flag if the latest charge is more than 5% above the average.
Error / alternative cases	Less than 3 months of data $\rightarrow$ results shown with a "low confidence" note. Irregular amounts (such as a utility bill) $\rightarrow$ shown with a "variable amount" label.

F06: TFSA / RRSP / FHSA contribution-room tracker

Field	Specification
Description	Computes the user's available contribution room for each registered account and warns before an over-contribution (which is taxed by CRA). Annual limits are loaded from a config file (<code>contribution_limits.json</code>), so updating them each year needs no code change.
User interaction (GUI)	<i>Planning</i> $\rightarrow$ <i>Registered accounts</i> : room gauge per account, contribution/withdrawal log, Check contribution box. CLI: <code>maplecfo room TFSA</code> , <code>maplecfo room check TFSA 3000</code> .
Input	<code>UserProfile</code> (birth year, year of Canadian residency, RRSP deduction limit from the CRA Notice of Assessment, FHSA opening year); logged contributions and withdrawals; proposed amount.
Output	<code>ContributionRoom</code> (limit, used, available) and <code>ContributionCheck</code> (OK / OVER with the excess / UNKNOWN with the missing fields).
AI involvement	Deterministic.
Expected workflow	TFSA : room = sum of annual limits from the first eligible year (the latest of 2009, the year the user turned 18, and the year they became resident), plus withdrawals made in previous years, minus contributions. FHSA : annual limit plus carry-forward (capped), lifetime limit. RRSP : the deduction limit entered by the user minus contributions. If the amount exceeds the room $\rightarrow$ <code>CONTRIBUTION_WARNING</code> event.
Error / alternative cases	Profile incomplete $\rightarrow$ UNKNOWN, with the missing fields listed. Limits file missing or corrupt $\rightarrow$ the feature is disabled with an explanatory message. Negative amount $\rightarrow$ validation error.

F07: FIRE (financial independence) projection

Field	Specification
Description	Projects when the user could reach financial independence: the year their invested assets reach annual spending divided by the chosen withdrawal rate. It compares scenarios at different savings rates.
User interaction (GUI)	<i>Planning</i> → FIRE form (current investments, yearly spending, yearly savings, expected real return, withdrawal rate) with a growth chart and a scenario table. CLI: <code>maplecfo fire --spend 40000 --save 15000 --return 0.05</code> .
Input	FireInputs (invested amount, yearly saving, annual spending, real return, withdrawal rate, current age).
Output	FireProjection (target amount, years and age at FIRE, balance for each year, scenarios).
AI involvement	Deterministic.
Expected workflow	target = spending ÷ withdrawal rate. Simulate year by year: balance = balance × (1 + r) + saving, until balance ≥ target (at most 70 years). Repeat for ±5% savings-rate scenarios.
Error / alternative cases	Negative values, return rate outside −5%...15%, or withdrawal rate outside 2%...6% → validation errors. Target never reached within 70 years → "not reachable with these inputs" plus a suggestion.

F08: Debt payoff planner (avalanche vs snowball)

Field	Specification
Description	Builds a month-by-month payoff schedule for the user's debts (cards, student loan, car loan) using the avalanche method (highest APR first) or the snowball method (smallest balance first), and compares total interest and time to become debt-free.
User interaction (GUI)	<i>Planning</i> → Debt: list of debts, method toggle, extra monthly payment, timeline chart, comparison table. CLI: <code>maplecfo debt plan --method avalanche --extra 200</code> .
Input	Debts (balance, APR, minimum payment); method; extra monthly amount.
Output	PayoffPlan (months to debt-free, total interest, schedule, payoff date per debt).
AI involvement	Deterministic.
Expected workflow	DebtPayoffPlanner asks its PayoffStrategy for the order, then simulates each month: add interest, pay the minimums, put the extra on the first debt in the order, and when a debt is paid off, roll its payment into the next one.
Error / alternative cases	No debts → "debt-free" message. Minimum payment lower than the monthly interest (the balance never shrinks) → warning for that debt. APR outside 0–60% → validation error.

F09: Savings goals with forecasts

Field	Specification
Description	The user creates goals (emergency fund, laptop, down payment) with a target amount and deadline. The system tracks progress and forecasts whether the goal will be met, using the cash-flow forecast.
User interaction (GUI)	<i>Goals</i> tab: goal cards with progress rings, New goal , Add deposit . CLI: <code>maplecfo goal add "Laptop" 1800 2027-04-30</code> , <code>maplecfo goal list</code> .
Input	Name, target, deadline, optional linked account; deposits.
Output	GoalForecast (required monthly saving, projected completion date, on-track flag).
AI involvement	Deterministic (the agent can <i>propose</i> a goal, which the user must confirm; see F10 and F11).
Expected workflow	Creation runs as a CreateGoalCommand (undoable). Forecast: required per month = (target – saved) ÷ months left, compared with the average monthly surplus from CashFlowForecaster.

Field	Specification
Error / alternative cases	Target ≤ 0 or deadline in the past → validation error. Surplus is negative → "not on track" with the shortfall. Goal reached → GOAL_UPDATED event and a congratulation banner.

F10: AI CFO chat (multi-step, tool-using agent with memory)

Field	Specification
Description	A conversational agent that answers free-form money questions by planning and calling finance tools (net worth, spending summary, budget status, contribution room, FIRE, debt plan, goals, recurring charges, affordability, propose action). It keeps conversation memory for follow-ups, quotes only tool-provided numbers, and can propose changes that the user must confirm.
User interaction (GUI)	Chat tab: message box, answer bubbles with "tools used" chips, Confirm / Dismiss buttons for proposed actions. CLI: <code>maplecfo ask "..."</code> , or an interactive <code>maplecfo chat</code> session.
Input	Natural-language message; conversation history; the user's financial data (only through tools).
Output	<code>AgentResponse</code> (answer, tools used, grounded flag, optional pending action id).
AI involvement	AI.
Expected workflow	Save the message in memory → build the prompt (system rules, a short profile summary, recent turns) → loop of at most 6 steps: call the LLM with the tool schemas → if it asks for tools, validate and run them and add the results to the conversation → if it gives a final answer, <code>ResponseValidator</code> checks every dollar amount against the tool results (if some are ungrounded, ask for one revision) and appends an educational disclaimer.
Error / alternative cases	LLM API error → one retry with backoff, then a "AI unavailable" message pointing to the manual tabs. Unknown tool or invalid arguments → error result sent back to the LLM, which must recover. Step limit reached → partial answer. Question out of scope (e.g. "which stock should I buy?") → polite refusal with education only. Ambiguous question → the agent asks one clarifying question.

F11: "Can I afford it?" affordability check

Field	Specification
Description	Given an amount and a date, decides whether a purchase is affordable, affordable with trade-offs, or not affordable . It takes the projected balance, committed savings goals, budget status and a safety buffer into account. Available as a form and as an agent tool.
User interaction (GUI)	Goals → <i>Can I afford it?</i> form, or ask in Chat. CLI: <code>maplecfo afford 2000 --on 2026-12-20</code> .
Input	Amount, target date.
Output	<code>AffordabilityResult</code> (verdict, headroom, shortfall, affected goals). In chat, an explanation plus an optional goal proposal.
AI involvement	Hybrid: the verdict is computed deterministically. The agent explains it and may propose a goal.
Expected workflow	<code>CashFlowForecaster</code> projects the balance up to the date (average income minus spending plus known recurring charges) → subtract the monthly contributions to committed goals and a safety buffer (1 month of fixed costs) → classify the result.
Error / alternative cases	Date in the past → validation error. Less than 2 months of history → "insufficient data" with low confidence. Amount ≤ 0 → validation error.

F12: Monthly summary with AI narrative

Field	Specification
Description	Produces a monthly summary (income, spending per category, budget results, net-worth change, subscriptions, goal progress) with a short AI-written narrative based only on the computed figures. It is shown on screen in the GUI and printed as formatted text in the CLI.
User interaction (GUI)	<i>Reports</i> tab: month picker, Generate , summary screen with charts. CLI: <code>maplecfo report 2026-09</code> .
Input	Month.
Output	<code>MonthlyReport</code> (figures + narrative) displayed in the GUI or CLI.
AI involvement	Hybrid : the figures are deterministic, the narrative is written by the LLM.
Expected workflow	<code>ReportService</code> gathers the figures from the services → <code>ReportSummarizer</code> sends <i>only the computed figures</i> to the LLM to write the narrative → the summary is displayed.
Error / alternative cases	No data for the month → message. LLM unavailable → template-based narrative (no AI).

3. UML class diagram

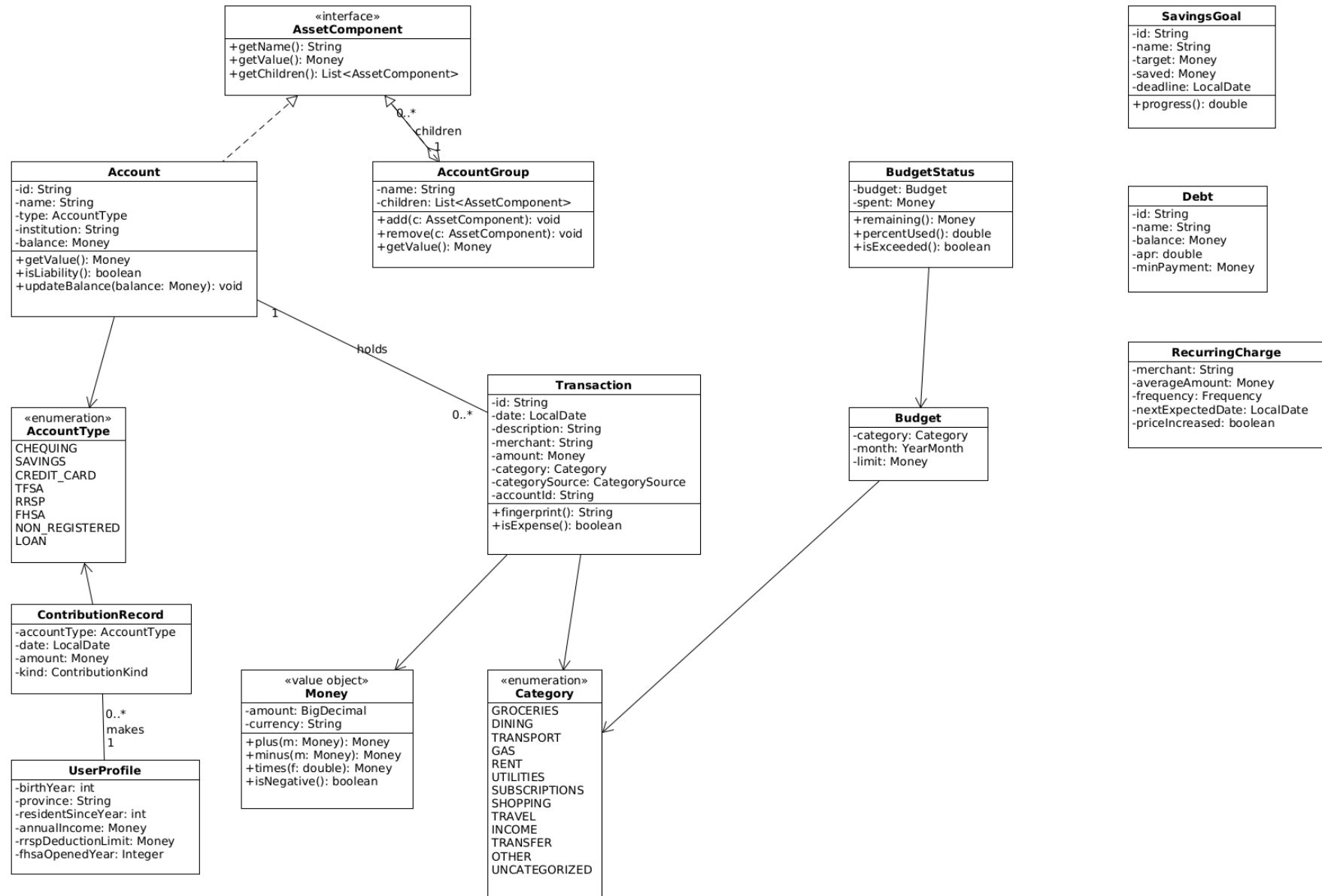
All diagrams are drawn in **UMLet**, and the editable `.uxf` file for each one is in `docs/stage1/diagrams/uxf/`. The complete class diagram is too large to read as a single picture, so it is presented in **six parts, one per package**. Classes that appear in more than one part (for example `BudgetService`, `LlmClient`) are the same class. Each part only repeats the members relevant to that part. The architecture diagram in 1.7 shows how the packages depend on each other.

Notation: `+` public, `-` private, `#` protected, *italic* / `*` abstract, underlined / `$` static. Solid arrow = association, dashed arrow = dependency, hollow triangle = inheritance/realisation, filled diamond = composition, hollow diamond = aggregation. Multiplicities are shown where they carry meaning.

3.1 Part 1: Domain model

Plain domain objects plus the **Composite** structure used for net worth. `Money` is an immutable value object (`BigDecimal`) so that no floating-point rounding errors reach financial totals.

Class diagram (1/6): Domain model (Composite: net worth)

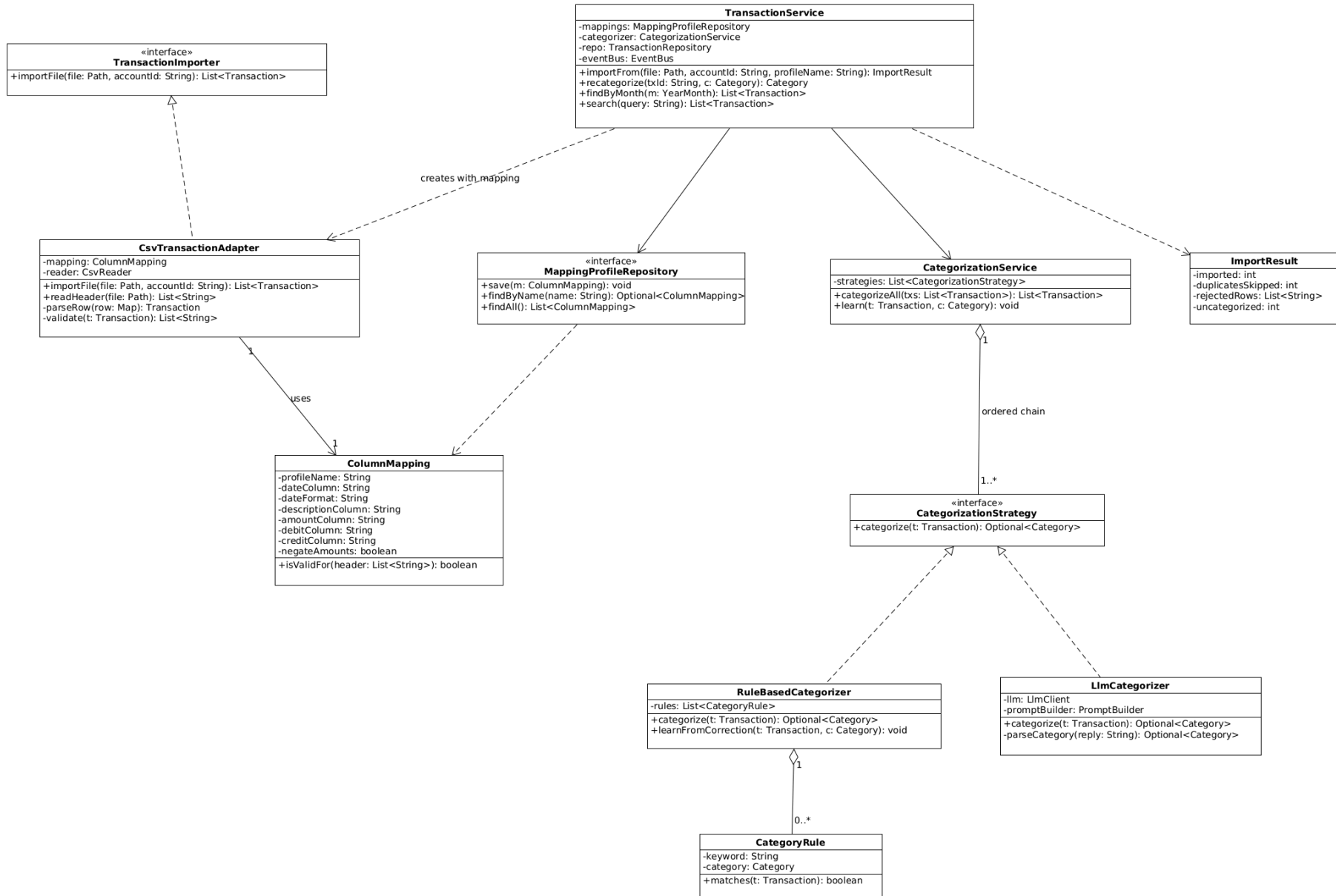


Class diagram (1/6): Domain model (Composite: net worth)

3.2 Part 2: Import and categorization

A single `CsvTransactionAdapter` **adapts** any bank's CSV layout to the `TransactionImporter` interface using a saved `ColumnMapping` profile, plus the **Strategy**-based categorization chain.

Class diagram (2/6): Import and categorization (Adapter, Strategy)

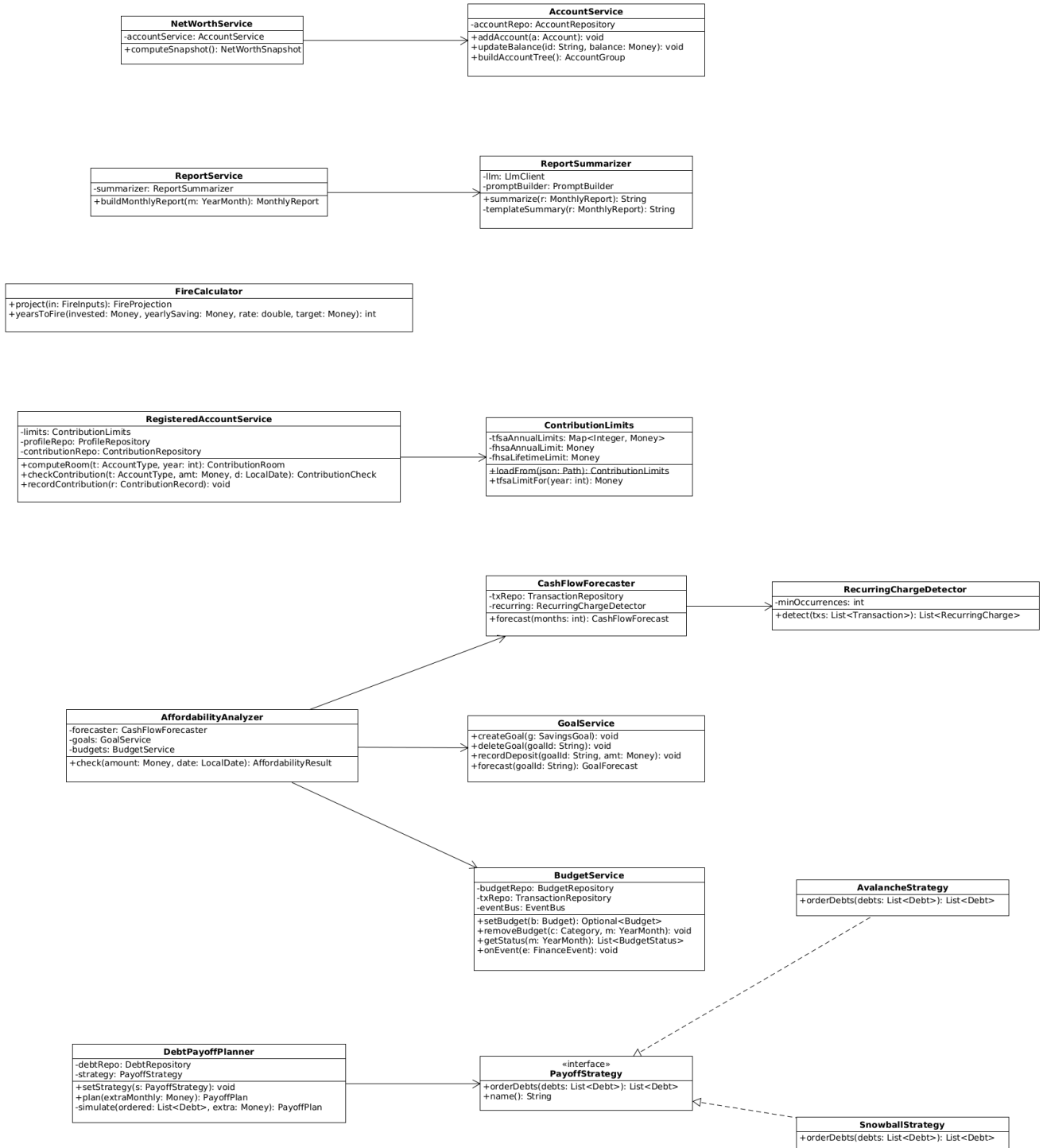


Class diagram (2/6): Import and categorization (Adapter, Strategy)

3.3 Part 3: Planning and analysis services

The deterministic "calculators". `DebtPayoffPlanner` uses a **Strategy** (`PayoffStrategy`). All of these classes are pure logic with injected repositories, which makes them easy to unit-test in Stage 3.

Class diagram (3/6): Planning and analysis services (Strategy)

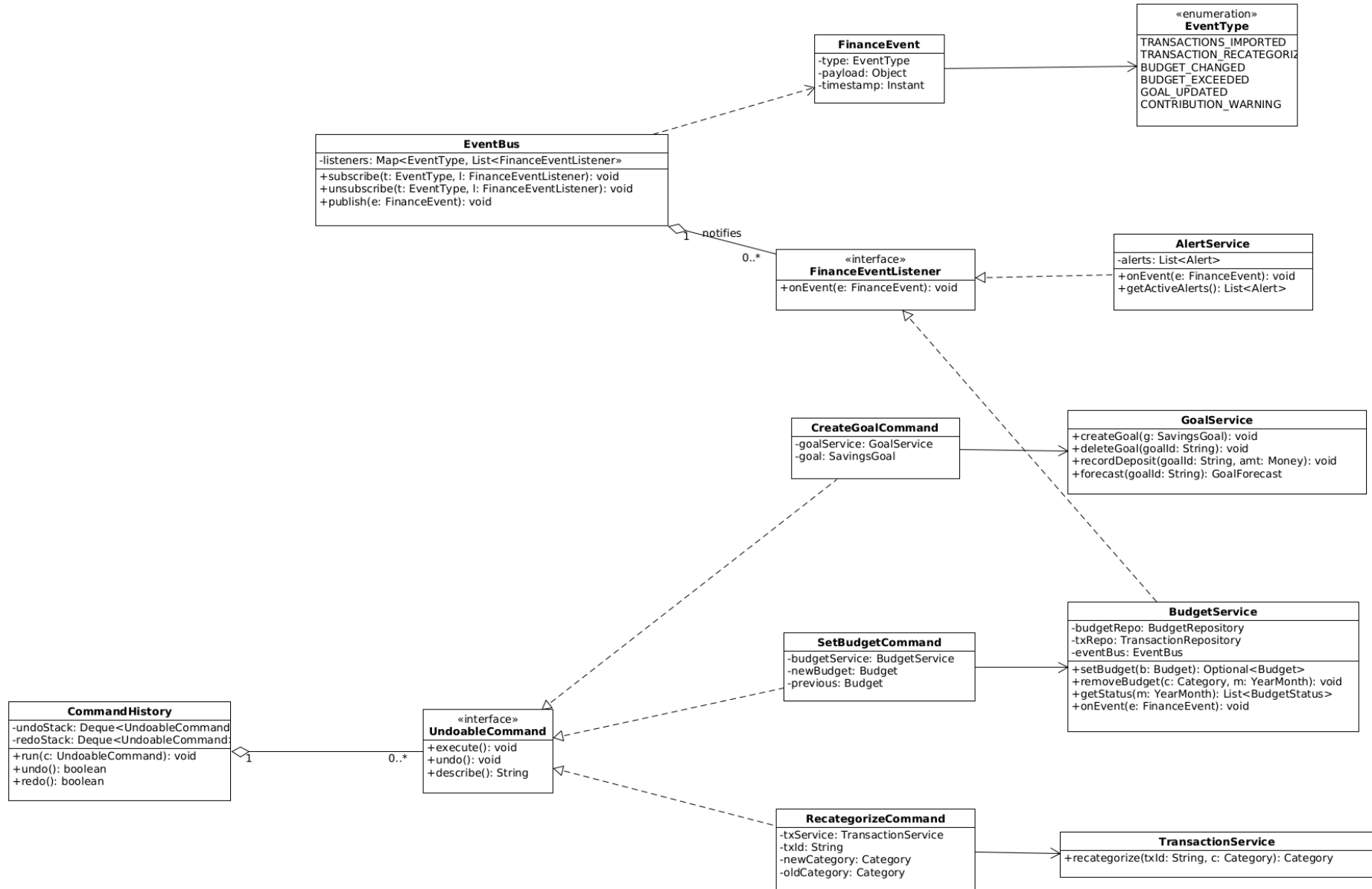


Class diagram (3/6): Planning and analysis services (Strategy)

3.4 Part 4: Events and undoable commands

The **Observer** event bus that keeps views, budgets and alerts in sync, and the **Command** objects that make user edits and agent-proposed actions undoable.

Class diagram (4/6): Events and undoable commands (Observer, Command)

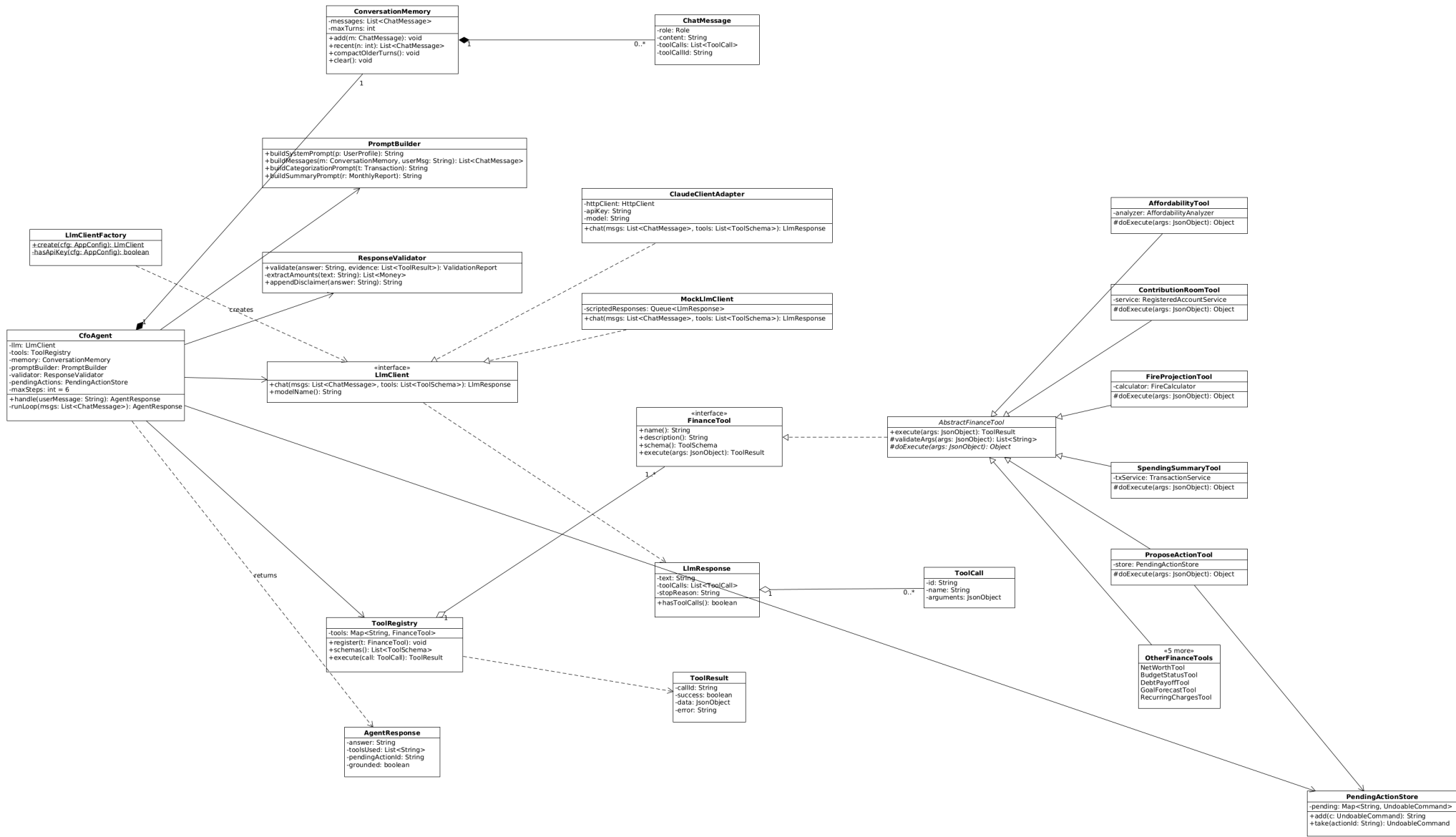


Class diagram (4/6): Events and undoable commands (Observer, Command)

3.5 Part 5: Agent layer (AI components)

The AI agent and its collaborators. `LlmClient` is an **Adapter** over the different LLM vendor APIs, `LlmClientFactory` is a **Factory Method** (Claude or offline mock, depending on configuration), `AbstractFinanceTool` is a **Template Method**, and `ProposeActionTool` produces **Command** objects.

Class diagram (5/6): Agent layer (Adapter, Factory Method, Template Method, Command)

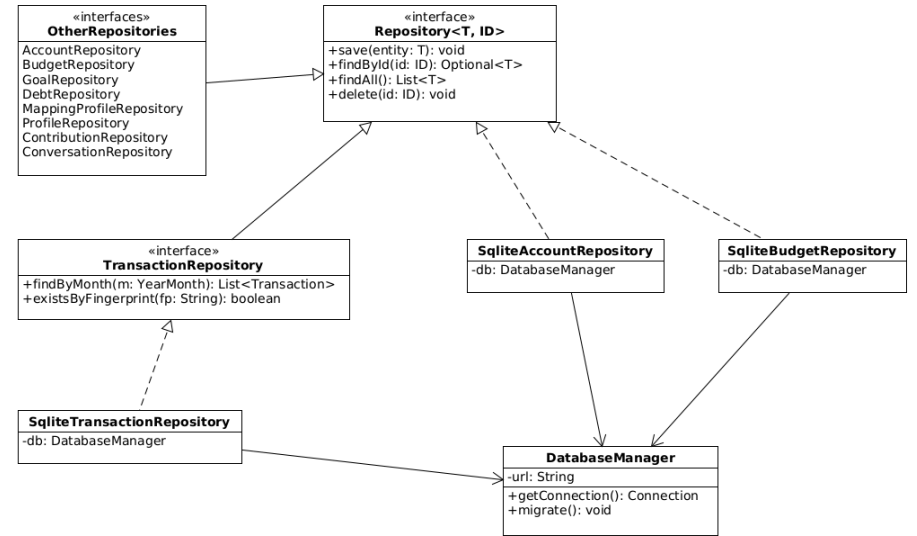
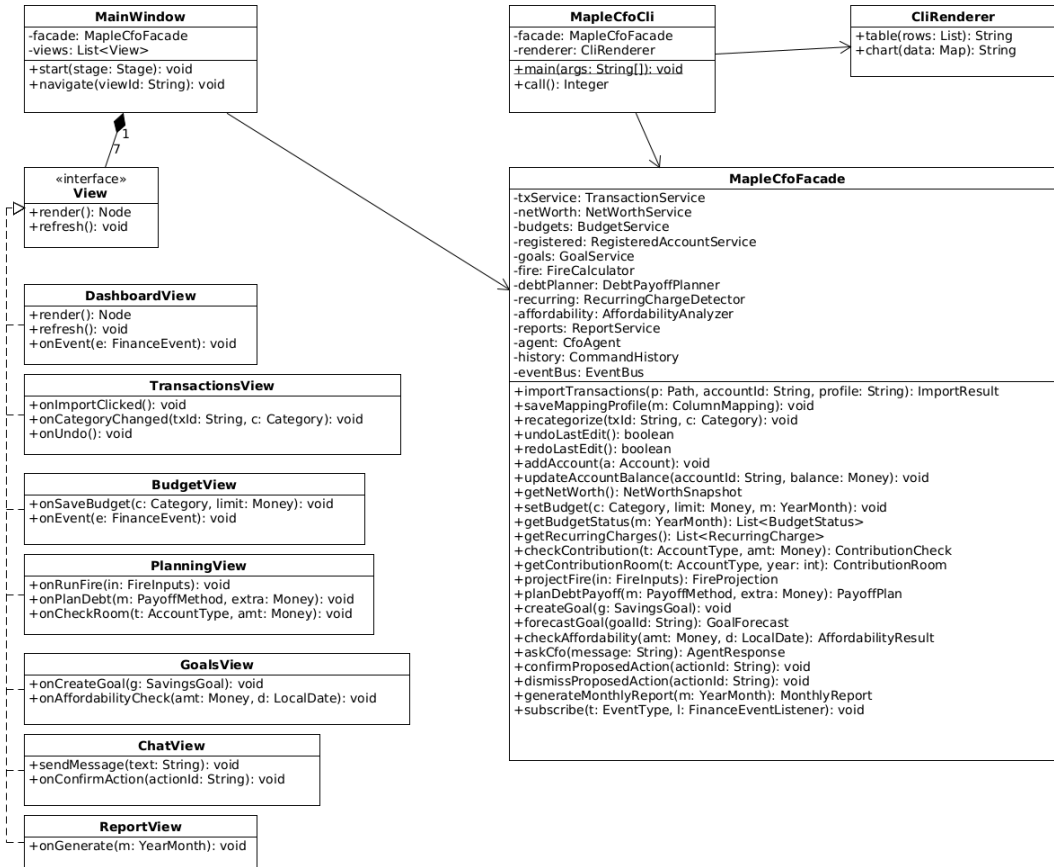


Class diagram (5/6): Agent layer (Adapter, Factory Method, Template Method, Command)

3.6 Part 6: Presentation, Facade and persistence

JavaFX views and the picocli CLI both depend only on the **Facade**. Persistence follows the **Repository/DAO** pattern over SQLite.

Class diagram (6/6): Presentation, Facade and persistence (Facade, MVC, Repository/DAO)



Class diagram (6/6): Presentation, Facade and persistence (Facade, MVC, Repository/DAO)

3.7 Key result / value classes (not drawn separately)

These are simple immutable data carriers (Java `records`) returned by services. They are named in the diagrams and tables: `ImportResult`, `NetWorthSnapshot`, `ContributionRoom`, `ContributionCheck`, `FireInputs`, `FireProjection`, `PayoffPlan`, `GoalForecast`, `CashFlowForecast`, `AffordabilityResult`, `MonthlyReport`, `ValidationReport`, `ToolSchema`, `Alert`, plus the enums `PayoffMethod`, `CategorySource`, `Frequency`, `ContributionKind`, `Role`.

4. Design pattern explanations

MapleCFO uses **eight GoF patterns** (at least five are required), plus the MVC and Repository architectural patterns. Each one solves a specific problem in this application.

4.1 Facade: `MapleCfoFacade`

Problem	Two user interfaces (GUI and CLI) need the same 20+ operations, which are spread across more than 12 services, the command history and the agent. Without a single entry point both UIs would depend on every subsystem and duplicate the coordination logic.
Participants and roles	<i>Facade</i> : <code>MapleCfoFacade</code> . <i>Subsystem classes</i> : <code>TransactionService</code> , <code>BudgetService</code> , <code>RegisteredAccountService</code> , <code>FireCalculator</code> , <code>DebtPayoffPlanner</code> , <code>GoalService</code> , <code>AffordabilityAnalyzer</code> , <code>ReportService</code> , <code>CfoAgent</code> , <code>CommandHistory</code> , <code>EventBus</code> . <i>Clients</i> : <code>MainWindow</code> and its views, <code>MapleCfoCli</code> .
Why appropriate	It gives low coupling between presentation and logic, one place to wrap operations in commands, and it guarantees that the GUI and the CLI behave identically. It is also the natural target for integration tests.
Harder without it	Every view and every CLI command would need references to many services and would have to remember to wrap edits in commands. Adding the CLI would mean duplicating orchestration code, and GUI and CLI behaviour would drift apart.

4.2 Adapter: `CsvTransactionAdapter` and `ClaudeClientAdapter`

Problem	(a) Every bank exports CSV with different column names, date formats and sign conventions, but the system needs uniform <code>Transaction</code> objects. (b) The Claude API has its own HTTP request format and tool-calling JSON, but the agent should work with simple Java objects (<code>ChatMessage</code> , <code>ToolCall</code> , <code>LlmResponse</code>).
Participants and roles	(a) <i>Target</i> : <code>TransactionImporter</code> . <i>Adapter</i> : <code>CsvTransactionAdapter</code> , configured by a <code>ColumnMapping</code> . <i>Adaptee</i> : the bank-specific CSV layout. <i>Client</i> : <code>TransactionService</code> . (b) <i>Target</i> : <code>LlmClient</code> . <i>Adapter</i> : <code>ClaudeClientAdapter</code> (plus <code>MockLlmClient</code> , a scripted implementation of the same interface). <i>Adaptee</i> : the Anthropic Messages API. <i>Client</i> : <code>CfoAgent</code> , <code>LlmCategorizer</code> , <code>ReportSummarizer</code> .
Why appropriate	One data-driven adapter supports any bank without new code: the user just saves a new mapping profile. The agent depends on the abstraction <code>LlmClient</code> , not on Claude's JSON format (Dependency Inversion), which makes a mock LLM possible for deterministic tests and offline demos.
Harder without it	CSV parsing would need a class per bank. <code>CfoAgent</code> would contain vendor-specific JSON handling, and it could not be unit-tested without network calls and API costs.

4.3 Template Method: `AbstractFinanceTool.execute()`

Problem	All 10 agent tools must follow the same safety steps: validate the LLM-supplied arguments, run the action, and wrap the outcome (or any exception) in a <code>ToolResult</code> . Only the core action differs from tool to tool.
Participants and roles	<i>Abstract class</i> : <code>AbstractFinanceTool</code> , with the template method <code>execute()</code> and the hooks <code>validateArgs()</code> and <code>doExecute()</code> . <i>Concrete classes</i> : <code>AffordabilityTool</code> , <code>ContributionRoomTool</code> , <code>FireProjectionTool</code> , <code>SpendingSummaryTool</code> , <code>ProposeActionTool</code> , <code>NetWorthTool</code> , <code>BudgetStatusTool</code> , <code>DebtPayoffTool</code> , <code>GoalForecastTool</code> , <code>RecurringChargesTool</code> .
Why appropriate	The invariant steps protect against bad LLM arguments. They are written and tested once and cannot be skipped by a subclass.
Harder without it	Each of the 10 tools would repeat the validation and error handling. One forgotten check would let the LLM run a tool with invalid arguments or crash the agent loop with an uncaught exception.

4.4 Factory Method: `LlmClientFactory.create()`

Problem	Which <code>LlmClient</code> to create depends on configuration: the real <code>ClaudeClientAdapter</code> when an API key and model name are configured, or <code>MockLlmClient</code> for tests and offline demo mode. Clients should not use <code>new ClaudeClientAdapter(...)</code> themselves or read configuration.
Participants and roles	<i>Creator:</i> <code>LlmClientFactory</code> with <code>create(AppConfig)</code> . <i>Product:</i> <code>LlmClient</code> . <i>Concrete products:</i> <code>ClaudeClientAdapter</code> , <code>MockLlmClient</code> . <i>Clients:</i> application start-up code, which injects the product into <code>CfoAgent</code> , <code>LlmCategorizer</code> and <code>ReportSummarizer</code> .
Why appropriate	Creation logic (reading the API key, model name and timeouts, and choosing demo mode) lives in one place, and the three AI components depend only on the product interface.
Harder without it	Configuration parsing and <code>if (apiKey == null)</code> checks would be copied into every AI component, and switching to demo mode or adding another provider would mean editing all of them.

4.5 Strategy: categorization and debt payoff

Problem	Several behaviours have interchangeable algorithms chosen at run time: <i>how to categorise</i> (rules or LLM) and <i>which debt to pay first</i> (avalanche or snowball).
Participants and roles	<i>Contexts:</i> <code>CategorizationService</code> , <code>DebtPayoffPlanner</code> . <i>Strategy interfaces:</i> <code>CategorizationStrategy</code> , <code>PayoffStrategy</code> . <i>Concrete strategies:</i> <code>RuleBasedCategorizer</code> , <code>LlmCategorizer</code> ; <code>AvalancheStrategy</code> , <code>SnowballStrategy</code> .
Why appropriate	The user switches payoff methods with a toggle and compares them. Categorization strategies are combined as an ordered chain (cheap and deterministic first, LLM last). Each strategy is small and testable on its own.
Harder without it	<code>DebtPayoffPlanner.plan()</code> would contain <code>if (method == AVALANCHE) ... else ...</code> branches inside the simulation loop. Adding a new method (for example "highest interest-to-balance ratio") would mean modifying tested code.

4.6 Observer: `EventBus` and `FinanceEventListener`

Problem	When transactions are imported or recategorised, many independent parts must react: budget status, alerts, the dashboard, the budget view. The code that imports transactions should not know about any of them.
Participants and roles	<i>Subject:</i> <code>EventBus</code> (<code>subscribe</code> , <code>unsubscribe</code> , <code>publish</code>). <i>Observer interface:</i> <code>FinanceEventListener</code> . <i>Concrete observers:</i> <code>BudgetService</code> , <code>AlertService</code> , <code>DashboardView</code> , <code>BudgetView</code> . <i>Publishers:</i> <code>TransactionService</code> , <code>BudgetService</code> , <code>GoalService</code> , <code>RegisteredAccountService</code> . <i>Event:</i> <code>FinanceEvent</code> with <code>EventType</code> .
Why appropriate	It keeps the GUI up to date without polling and decouples the services from each other. The same events drive the CLI's alert output.
Harder without it	<code>TransactionService</code> would have to call <code>budgetService.recheck()</code> , <code>dashboard.refresh()</code> and so on directly, creating circular dependencies between the domain and the UI. Every new view would mean editing the services.

4.7 Command: `UndoableCommand`, `CommandHistory` and `PendingActionStore`

Problem	(a) Users need undo/redo for edits (recategorise, set budget, create goal). (b) The agent must never modify data by itself . Its proposed actions need to be stored as objects, shown to the user, and executed only after confirmation.
Participants and roles	<i>Command interface:</i> <code>UndoableCommand</code> (<code>execute</code> , <code>undo</code> , <code>describe</code>). <i>Concrete commands:</i> <code>RecategorizeCommand</code> , <code>SetBudgetCommand</code> , <code>CreateGoalCommand</code> . <i>Receivers:</i> <code>TransactionService</code> , <code>BudgetService</code> , <code>GoalService</code> . <i>Invoker:</i> <code>CommandHistory</code> (undo/redo stacks). <i>Client:</i> <code>MapleCfoFacade</code> ; <code>ProposeActionTool</code> creates commands and stores them in <code>PendingActionStore</code> until the user confirms.

Why appropriate	Requests become objects, which gives undo/redo, a readable description for the Confirm dialog (<code>describe()</code>), and a human-in-the-loop safety guarantee for AI actions.
Harder without it	Each edit would need its own ad-hoc undo code, and there would be no clean way to represent "an action the AI suggested but the user has not approved yet".

4.8 Composite: `AssetComponent`, `Account` and `AccountGroup`

Problem	Net worth is a tree: groups (Banking, Registered, Debts) contain accounts, and groups can contain sub-groups (for example Registered → TFSA / RRSP / FHSA). Clients want to ask any node "what is your value?" in the same way.
Participants and roles	<i>Component</i> : <code>AssetComponent</code> . <i>Leaf</i> : <code>Account</code> (negative value if it is a liability). <i>Composite</i> : <code>AccountGroup</code> (adds up its children). <i>Client</i> : <code>NetWorthService</code> , the dashboard chart.
Why appropriate	The dashboard breakdown, the subtotals and the net-worth total all come from the same recursive <code>getValue()</code> , and new grouping levels need no code changes.
Harder without it	The dashboard and <code>NetWorthService</code> would need type checks and nested loops for each level, and subtotals would be computed in several places.

4.9 Architectural patterns (in addition to the eight above)

- **MVC**: *Model* = domain objects and services. *View* = the JavaFX `View` classes (and `CliRenderer` for text). *Controller* = the views' event handlers delegating to `MapleCfoFacade`. Views never contain financial logic.
- **Repository / DAO**: `Repository<T, ID>` with SQLite implementations isolates SQL. Services can be tested with in-memory fakes.

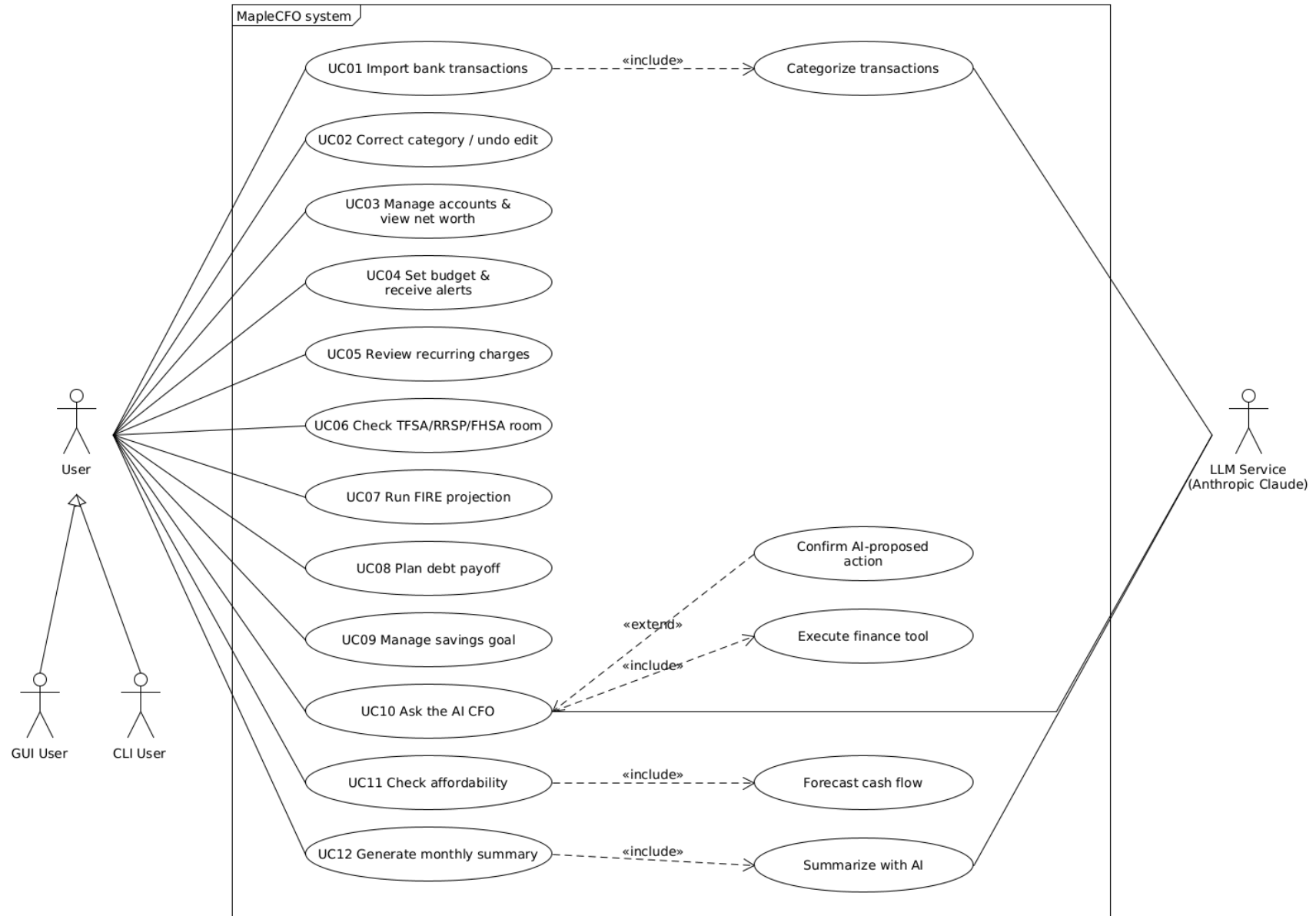
5. Use-case diagram

Actors

Actor	Type	Description
User	Primary	A student or young professional managing their money. Specialised into GUI User and CLI User (both reach the same use cases).
LLM Service	Secondary (external AI service)	The Anthropic Claude API, reached through <code>LlmClient</code> . Takes part in categorisation, the AI CFO chat and report summaries.

The 12 main use cases cover all 12 features. `«include»` marks behaviour that always happens as part of a use case. `«extend»` marks optional behaviour (confirming an action the AI proposed).

Use-case diagram: MapleCFO



Use-case diagram: MapleCFO

Notation: stick-figure actors, use-case ovals inside the system boundary, and dashed «include»/«extend» arrows. All UML diagrams in this report are drawn in **UMLet**; the editable `.uxf` files are in `docs/stage1/diagrams/uxf/`.

6. Detailed use-case descriptions

UC01: Import bank transactions

Actors	User (primary); LLM Service (secondary, through the included "Categorize transactions")
Goal	Bring the user's bank or card transactions into MapleCFO, categorised and free of duplicates.
Preconditions	The application is running; at least one account exists; the user has a CSV export from their bank.
Trigger	The user clicks Import CSV (GUI) or runs <code>maplecfo import <file></code> (CLI).
Main success scenario	1. The user selects a CSV file, the target account and a saved mapping profile. 2. The system checks that the profile matches the file's header. 3. The system reads, parses and validates each row. 4. The system removes duplicates of transactions already stored. 5. The system categorises the transactions (rules first, then AI) (<i>include: Categorize transactions</i>). 6. The system saves the transactions and notifies budgets and views. 7. The system shows a summary (imported, duplicates, rejected rows, need review).
Alternative / exception flows	1a/2a. No profile yet, or the profile does not match → the system shows the column-mapping dialog; the user maps date, description and amount columns and saves a new profile; continue at 3. 3a. File unreadable or empty → error message, nothing saved, use case ends. 3b. Some rows invalid → valid rows continue, invalid rows are listed with reasons. 5a. LLM unavailable or invalid answer → affected transactions are marked "needs review". 6a. A budget is exceeded → an alert is shown (see UC04).
Postconditions	New transactions are stored and categorised (or flagged); budgets and dashboard reflect them; the import is summarised; a new mapping profile is saved if one was created.
Related features	F01, F02, F04

UC02: Correct a category / undo an edit

Actors	User
Goal	Fix a wrong category (and teach the system), and be able to undo or redo edits.
Preconditions	At least one transaction exists.
Trigger	The user changes a category in the table, or presses Undo/Redo.
Main success scenario	1. The user selects a new category for a transaction. 2. The system runs a recategorise command and stores the old category. 3. The system learns a merchant rule for future imports. 4. The system updates budgets and views. 5. The user presses Undo . 6. The system restores the previous category.
Alternative / exception flows	2a. The transaction no longer exists → error; nothing is added to the history. 5a. Nothing to undo → Undo is disabled or reports "nothing to undo". 5b. The user presses Redo → the command is executed again.
Postconditions	The transaction has the chosen category; history stacks are consistent.
Related features	F02 (also F04 and F09, whose edits use the same undo mechanism)

UC03: Manage accounts and view net worth

Actors	User
Goal	Keep account balances up to date and see total net worth and its breakdown.
Preconditions	The application is running.
Trigger	The user opens the Dashboard or edits an account balance.
Main success scenario	1. The user adds or edits an account (type, institution, balance). 2. The system validates and saves it. 3. The system builds the account tree and computes group subtotals and net worth. 4. The system displays the net-worth card, the breakdown chart and the history.

Alternative / exception flows	2a. Negative balance on an asset account → validation error. 3a. No accounts → empty state with a prompt to add one.
Postconditions	Balances are stored and a net-worth snapshot is recorded.
Related features	F03

UC04: Set a budget and receive alerts

Actors	User
Goal	Limit monthly spending per category and be warned when approaching or exceeding the limit.
Preconditions	The application is running (transactions optional).
Trigger	The user enters a limit on the Budget tab or runs <code>maplecfo budget set</code> .
Main success scenario	1. The user chooses a category and a monthly limit. 2. The system validates it and runs a set-budget command (undoable). 3. The system computes the current status of every budget. 4. When new transactions arrive, the system recomputes the status. 5. If a category reaches 80% or 100%, the system shows an alert.
Alternative / exception flows	2a. Limit ≤ 0 → validation error. 2b. A budget already exists → it is replaced (the previous one is kept for undo).
Postconditions	The budget is stored; status and alerts are up to date.
Related features	F04

UC05: Review recurring charges

Actors	User
Goal	See all subscriptions and recurring bills, their yearly cost and any price increases.
Preconditions	Transactions are imported (ideally at least 3 months).
Trigger	The user opens the Subscriptions panel or runs <code>maplecfo recurring</code> .
Main success scenario	1. The system loads the last 6 months of transactions. 2. The system detects regularly repeating merchants. 3. The system computes averages, the next expected dates and price increases. 4. The system shows the list and the yearly total.
Alternative / exception flows	1a. Less than 3 months of data → results with a low-confidence note. 3a. Variable amounts → labelled "variable".
Postconditions	None (read-only).
Related features	F05

UC06: Check TFSA / RRSP / FHSA contribution room

Actors	User
Goal	Know the available room and avoid over-contributing.
Preconditions	The user profile contains a birth year and residency year (and an RRSP limit and FHSA opening year where relevant).
Trigger	The user opens Registered Accounts or enters an amount to check.
Main success scenario	1. The user selects an account type and enters an amount. 2. The system loads the profile, the limits file and the contribution history. 3. The system computes the available room. 4. The system compares the amount with the room and shows OK and the room left.

Alternative / exception flows	2a. Profile incomplete → the system lists the missing fields. 2b. Limits file missing → the feature is unavailable with an explanation. 4a. Amount exceeds the room → a warning shows the excess, and a CONTRIBUTION_WARNING alert is raised.
Postconditions	Optionally, the user logs the contribution, which updates the history.
Related features	F06

UC07: Run a FIRE projection

Actors	User
Goal	Estimate when financial independence is reachable and compare savings scenarios.
Preconditions	None (values can be typed or pre-filled from the data).
Trigger	The user fills in the FIRE form and clicks Project .
Main success scenario	1. The user enters investments, spending, savings, expected return and withdrawal rate. 2. The system validates the inputs. 3. The system computes the target and simulates year by year. 4. The system shows the FIRE age, a growth chart and a scenario table.
Alternative / exception flows	2a. Invalid values → field errors. 3a. Target not reached within 70 years → "not reachable" with suggestions.
Postconditions	None (the scenario may optionally be saved).
Related features	F07

UC08: Plan debt payoff

Actors	User
Goal	Get a month-by-month plan to become debt-free and compare the avalanche and snowball methods.
Preconditions	At least one debt is entered.
Trigger	The user chooses a method and extra amount and clicks Plan .
Main success scenario	1. The user selects a method and an extra monthly payment. 2. The system orders the debts using the selected strategy. 3. The system simulates month by month. 4. The system shows the timeline, total interest and debt-free date, and a comparison with the other method.
Alternative / exception flows	1a. No debts → "debt-free" message. 3a. A minimum payment does not cover the interest → warning for that debt.
Postconditions	None.
Related features	F08

UC09: Manage a savings goal

Actors	User
Goal	Create a goal, record deposits and know whether the goal is on track.
Preconditions	None.
Trigger	The user clicks New goal or Add deposit .
Main success scenario	1. The user enters a name, target and deadline. 2. The system validates the goal and runs a create-goal command. 3. The system forecasts the required monthly saving using the cash-flow forecast. 4. The system shows the progress ring and the on-track status.
Alternative / exception flows	2a. Target ≤ 0 or deadline in the past → error. 3a. Negative surplus → "not on track" with the shortfall. 4a. Goal reached → a celebration banner.

Postconditions	The goal is stored (undoable).
Related features	F09

UC10: Ask the AI CFO

Actors	User (primary); LLM Service (secondary)
Goal	Get a personalised, data-backed answer to a free-form money question.
Preconditions	An LLM is configured (API key or local model); some financial data exists.
Trigger	The user sends a message in Chat or runs <code>maplecfo ask "..."</code> .
Main success scenario	1. The user asks a question. 2. The system adds it to the conversation memory and builds the prompt. 3. The LLM decides which tools it needs. 4. The system validates and runs the requested tools (<i>include: Execute finance tool</i>) and returns the results to the LLM. 5. Steps 3–4 repeat until the LLM produces a final answer (at most 6 steps). 6. The system checks that the answer's figures are backed by tool results and appends a disclaimer. 7. The system shows the answer and the tools used.
Alternative / exception flows	3a. LLM unavailable → one retry, then an "AI unavailable" message with links to the manual tabs. 4a. Invalid tool arguments → an error result is returned to the LLM so it can correct itself. 5a. Step limit reached → a partial answer is shown. 6a. Ungrounded figures → the LLM is asked to revise once. 7a. The LLM proposes an action → Confirm/Dismiss buttons appear (<i>extend: Confirm AI-proposed action</i>). 1a. Question out of scope (e.g. stock picks) → the agent declines and offers general education.
Postconditions	The answer is shown and stored in memory; no data is changed unless the user confirms a proposed action.
Related features	F10 (tools reach F03–F09, F11)

UC11: Check affordability

Actors	User
Goal	Find out whether a planned purchase fits the user's finances, and what the trade-offs are.
Preconditions	At least 2 months of transactions.
Trigger	The user fills in the "Can I afford it?" form, or asks in Chat.
Main success scenario	1. The user enters an amount and a date. 2. The system forecasts the balance up to the date (<i>include: Forecast cash flow</i>). 3. The system subtracts committed goal contributions and a safety buffer. 4. The system classifies the purchase as affordable, affordable with trade-off, or not affordable, and shows the headroom or shortfall. 5. (In chat) The agent explains the result and may propose a savings goal.
Alternative / exception flows	1a. Date in the past or amount ≤ 0 → validation error. 2a. Not enough history → "insufficient data". 5a. The user confirms the proposed goal → it is created through UC09's command.
Postconditions	None, unless a proposed goal is confirmed.
Related features	F11, F10, F09

UC12: Generate a monthly summary

Actors	User (primary); LLM Service (secondary)
Goal	Get a readable summary of a month's finances.
Preconditions	Transactions exist for the chosen month.
Trigger	The user picks a month and clicks Generate , or runs <code>maplecfo report <month></code> .
Main success scenario	1. The user selects a month. 2. The system computes the figures (income, spending, budgets, net worth, subscriptions, goals). 3. The system asks the LLM to write a short narrative from those figures (<i>include: Summarize with AI</i>). 4. The system displays the summary with charts (GUI) or as formatted text (CLI).

Alternative / exception flows	2a. No data for the month → message, use case ends. 3a. LLM unavailable → template narrative is used instead.
Postconditions	None (read-only).
Related features	F12

7. Sequence diagrams

Ten sequence diagrams cover every feature. Features with the same interaction structure share a diagram (for example FIRE and debt payoff). All participants and messages use the **class and method names from the class diagram**.

`alt/opt` fragments show error and alternative flows.

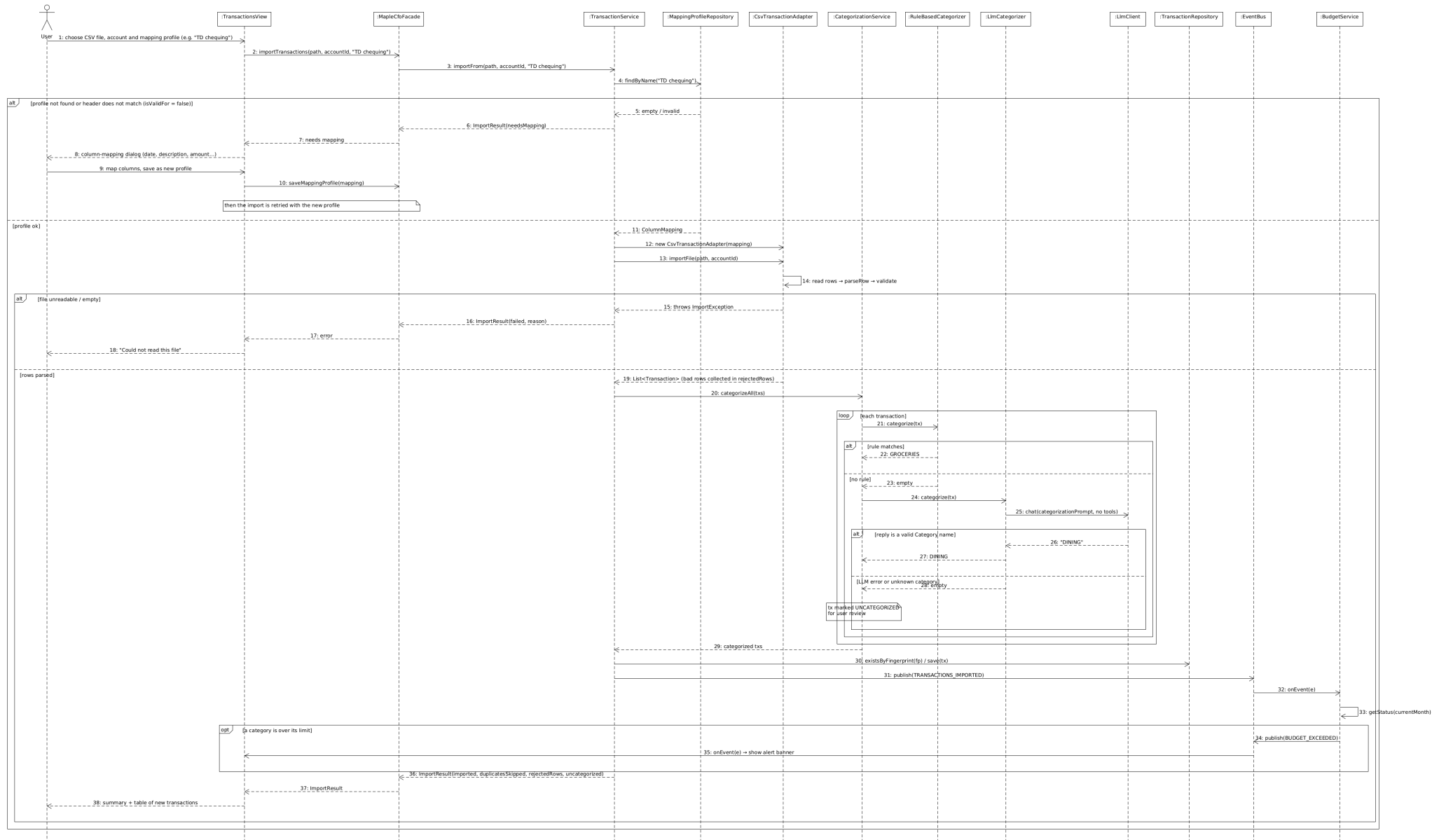
The GUI view is drawn as the boundary object. The **CLI follows exactly the same path**, because `MapleCfoCli` calls the same `MapleCfoFacade` methods (shown explicitly in SD08 and SD10).

SD	Title	Features	Use cases
SD01	Import and categorize transactions	F01, F02, F04	UC01, UC04
SD02	Correct a category / set a budget, with undo	F02, F04	UC02, UC04
SD03	Update an account and view net worth	F03	UC03
SD04	Check contribution room	F06	UC06
SD05	FIRE projection and debt payoff plan	F07, F08	UC07, UC08
SD06	Detect recurring charges	F05	UC05
SD07	Create and forecast a savings goal	F09	UC09
SD08	Ask the AI CFO (agent loop)	F10	UC10
SD09	"Can I afford it?" with a proposed action	F11, F10, F09	UC11, UC10
SD10	Generate the monthly summary	F12	UC12

SD01: Import and categorize transactions

Shows the mapping-profile lookup (with the column-mapping fallback), the `CsvTransactionAdapter` (Adapter), the Strategy chain for categorisation (rules first, then the LLM), and Observer notifications that trigger budget alerts.

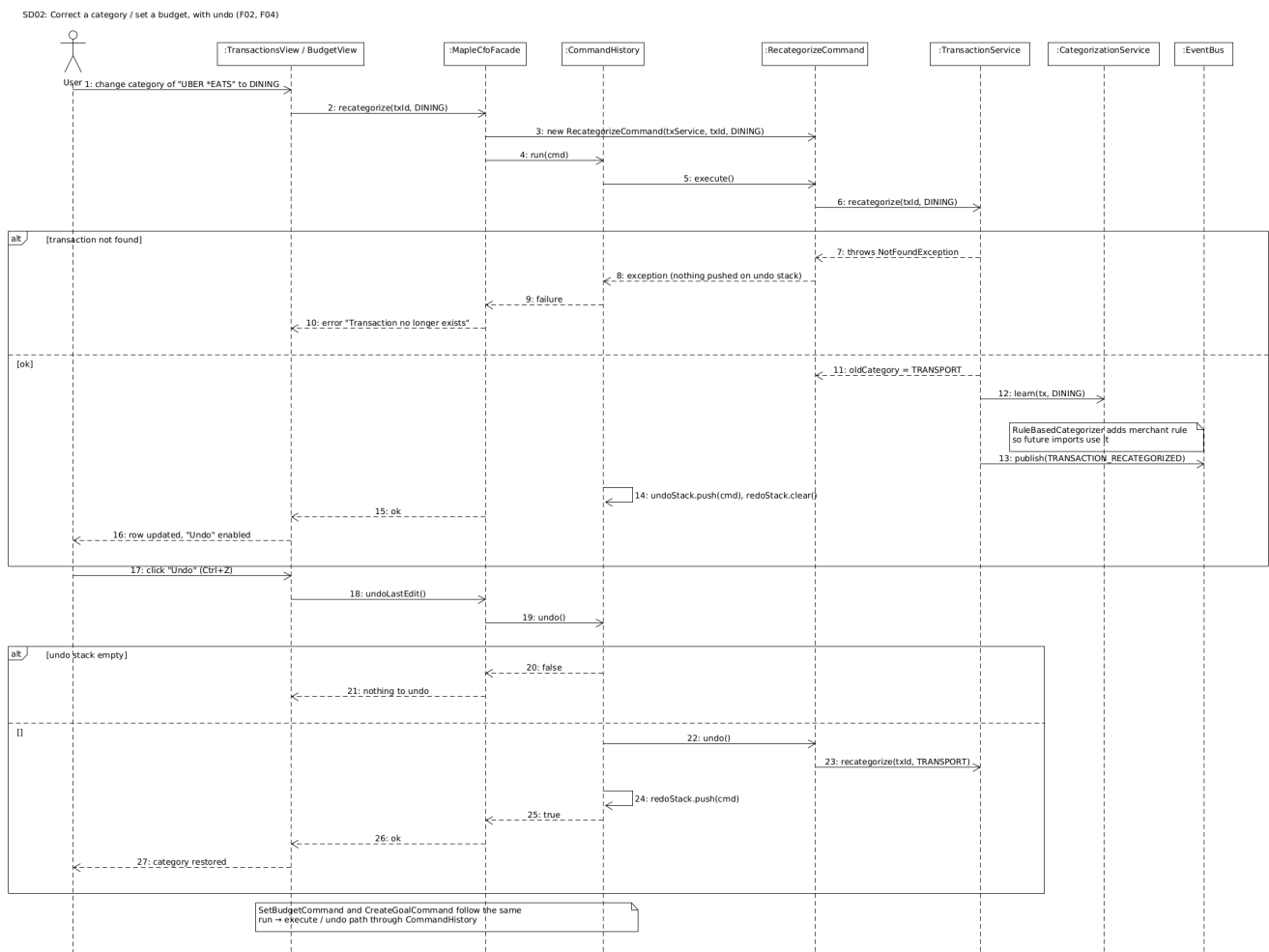
SD01: Import and categorize bank transactions (F01, F02, F04)



SD01: Import and categorize bank transactions (F01, F02, F04)

SD02: Correct a category / set a budget, with undo

Shows the Command pattern: the facade wraps the edit in a command, `CommandHistory` executes it and keeps it for undo/redo.

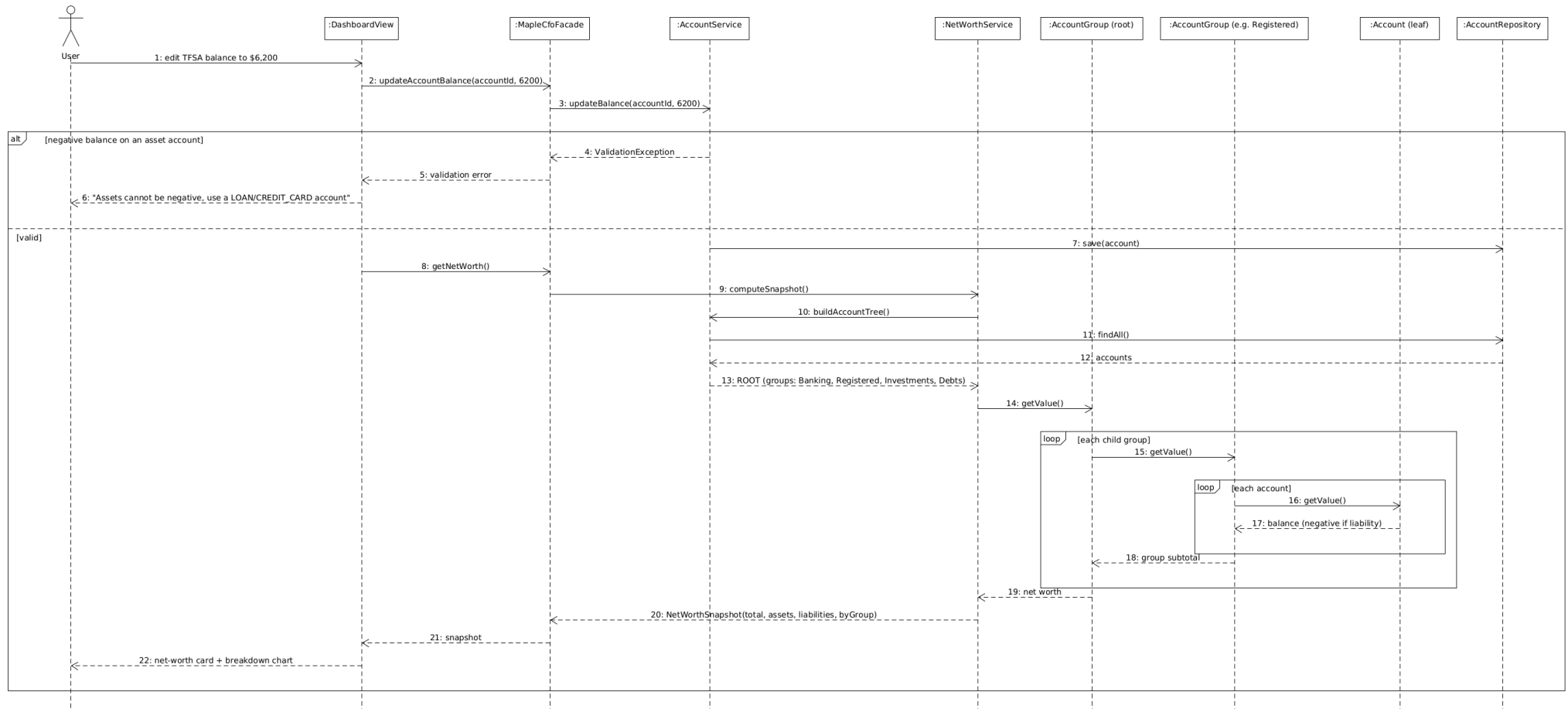


SD02: Correct a category / set a budget, with undo (F02, F04)

SD03: Update an account and view net worth

Shows the Composite recursion: the root `AccountGroup` adds up its groups, which add up their accounts.

SD03: Update an account and view net worth (F03)

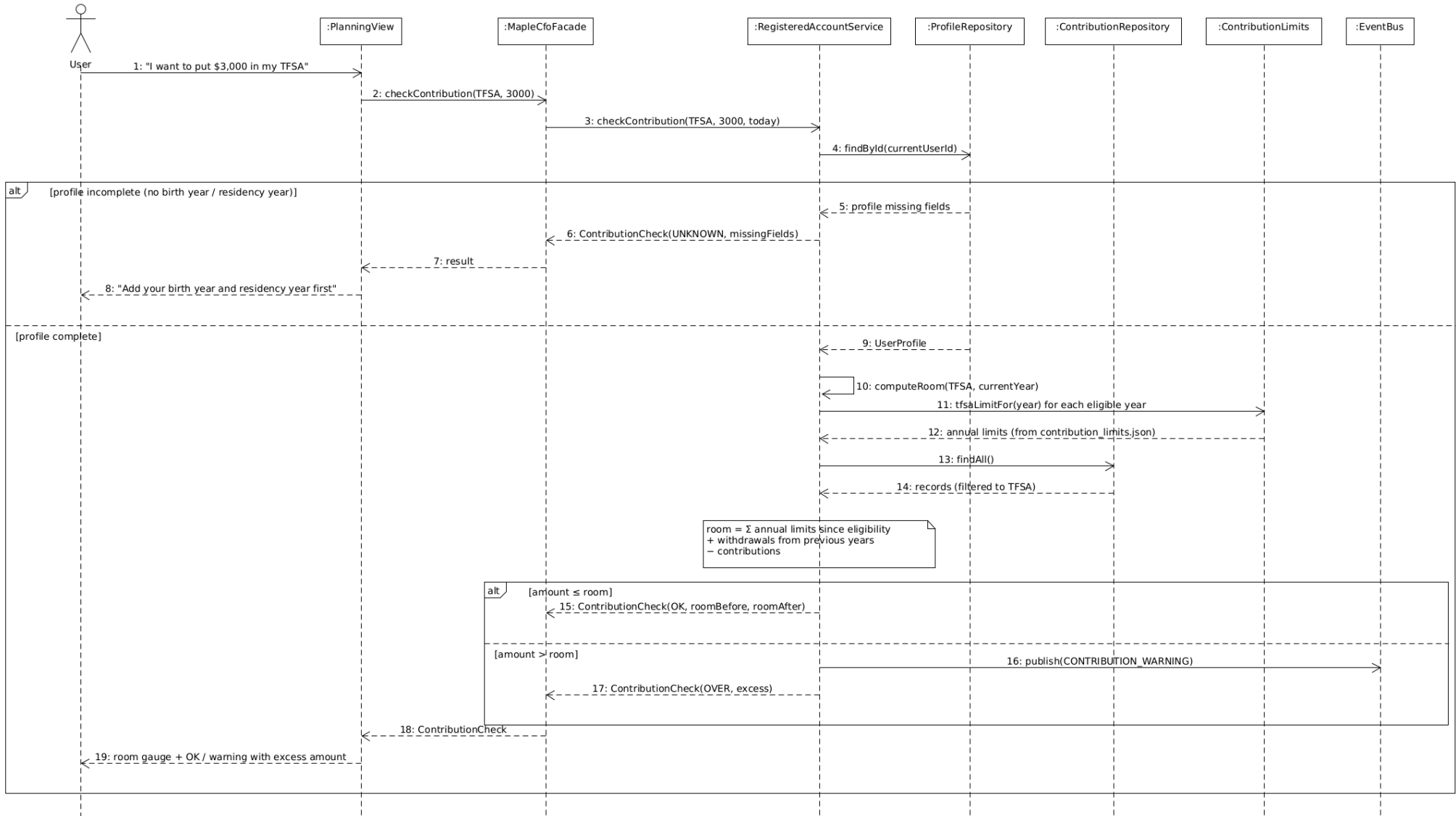


SD03: Update an account and view net worth (F03)

SD04: Check TFSA / RRSP / FHSA contribution room

Deterministic rule computation with configuration-driven limits and an over-contribution warning event.

SD04: Check TFSA / RRSP / FHSA contribution room (F06)

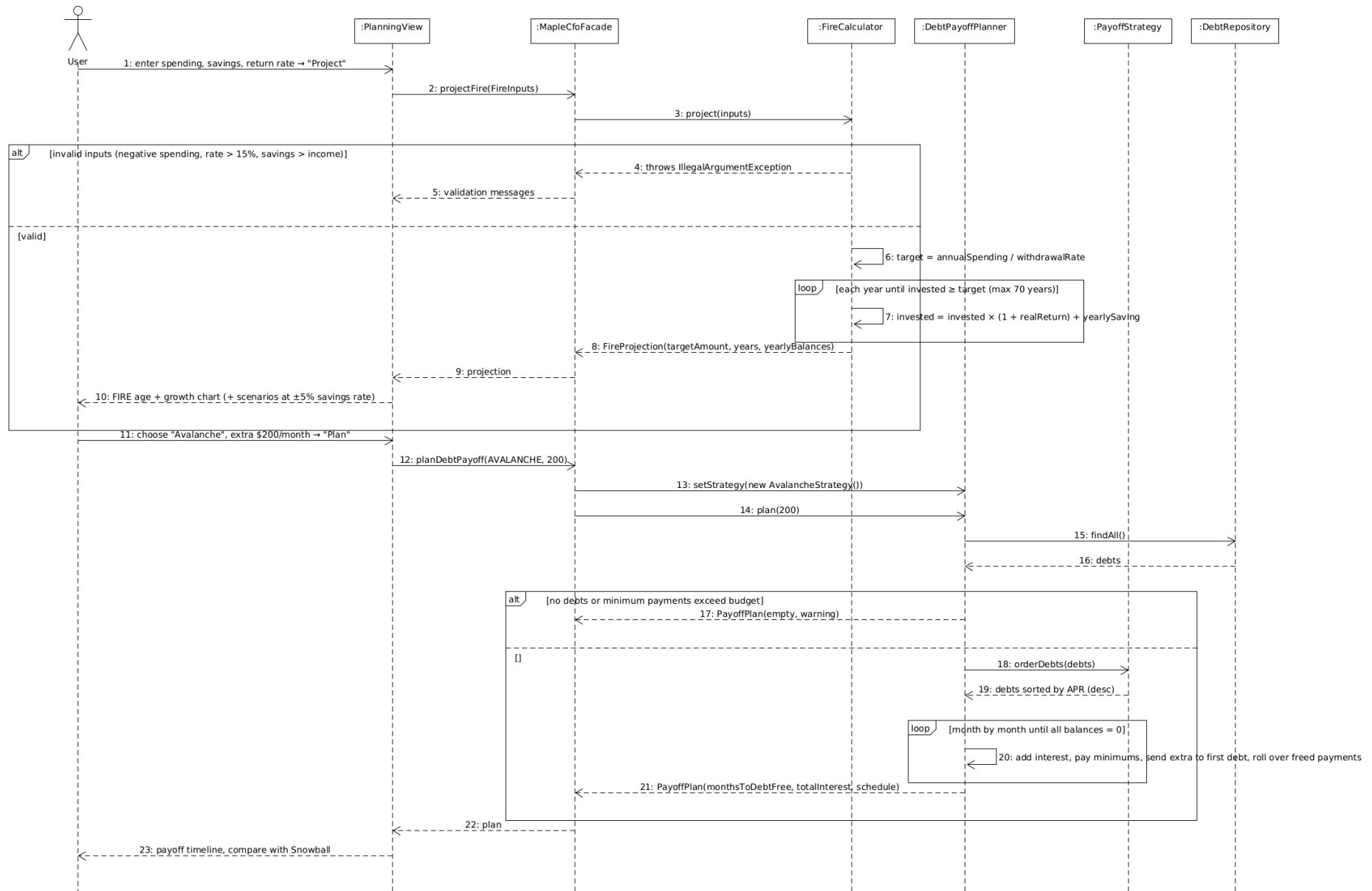


SD04: Check TFSA / RRSP / FHSA contribution room (F06)

SD05: FIRE projection and debt payoff plan

Two deterministic calculators. The debt planner uses the Strategy pattern, selected at run time.

SD05: FIRE projection and debt payoff plan (F07, F08)

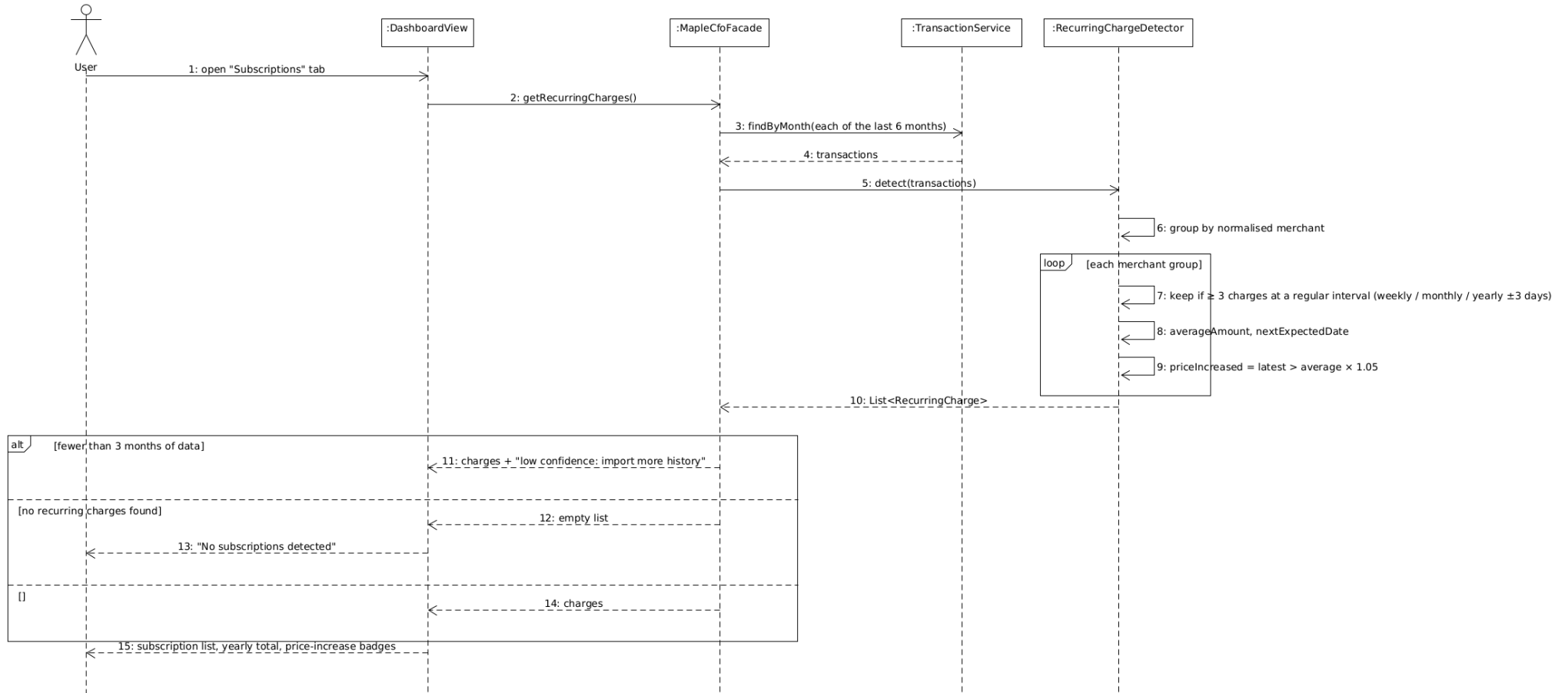


SD05: FIRE projection and debt payoff plan (F07, F08)

SD06: Detect recurring charges

A read-only analysis built on the transaction history.

SD06: Detect recurring charges and subscriptions (F05)

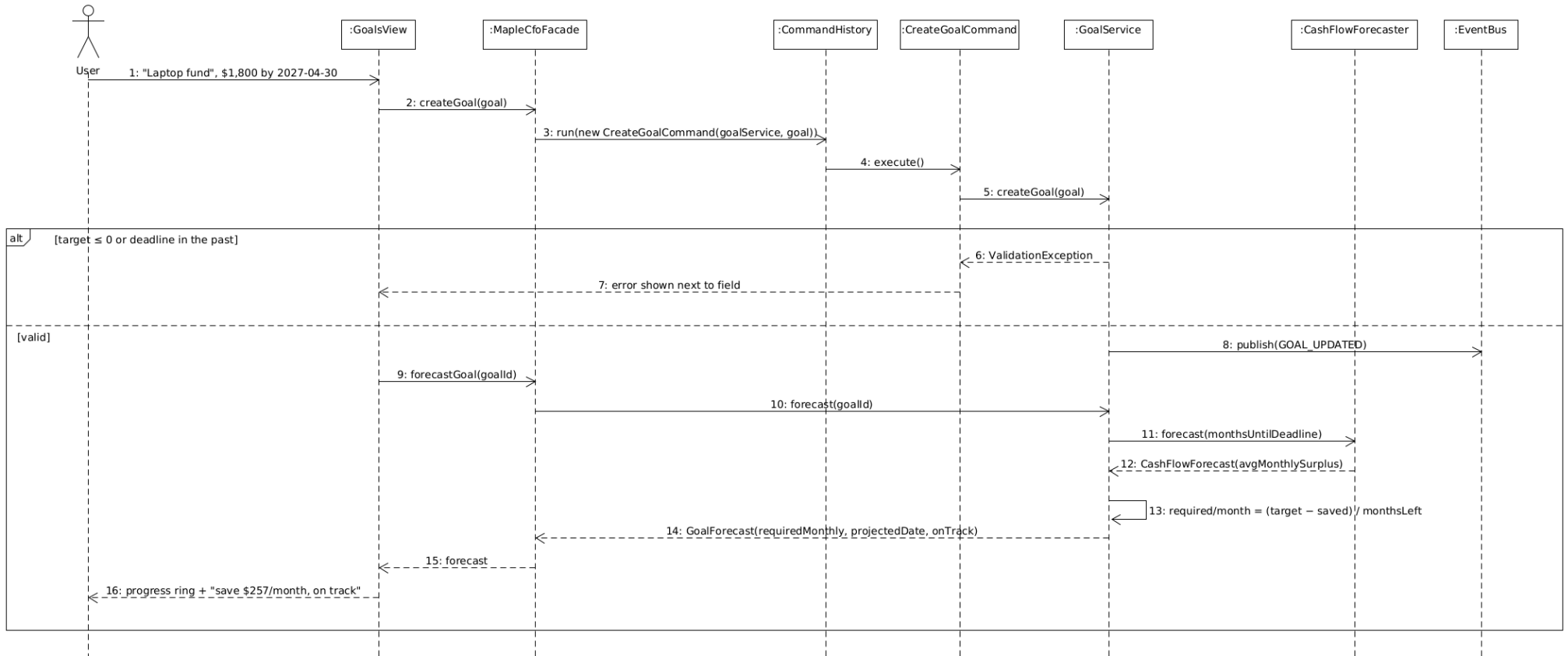


SD06: Detect recurring charges and subscriptions (F05)

SD07: Create and forecast a savings goal

Goal creation as an undoable command, followed by a forecast that uses the cash-flow forecaster.

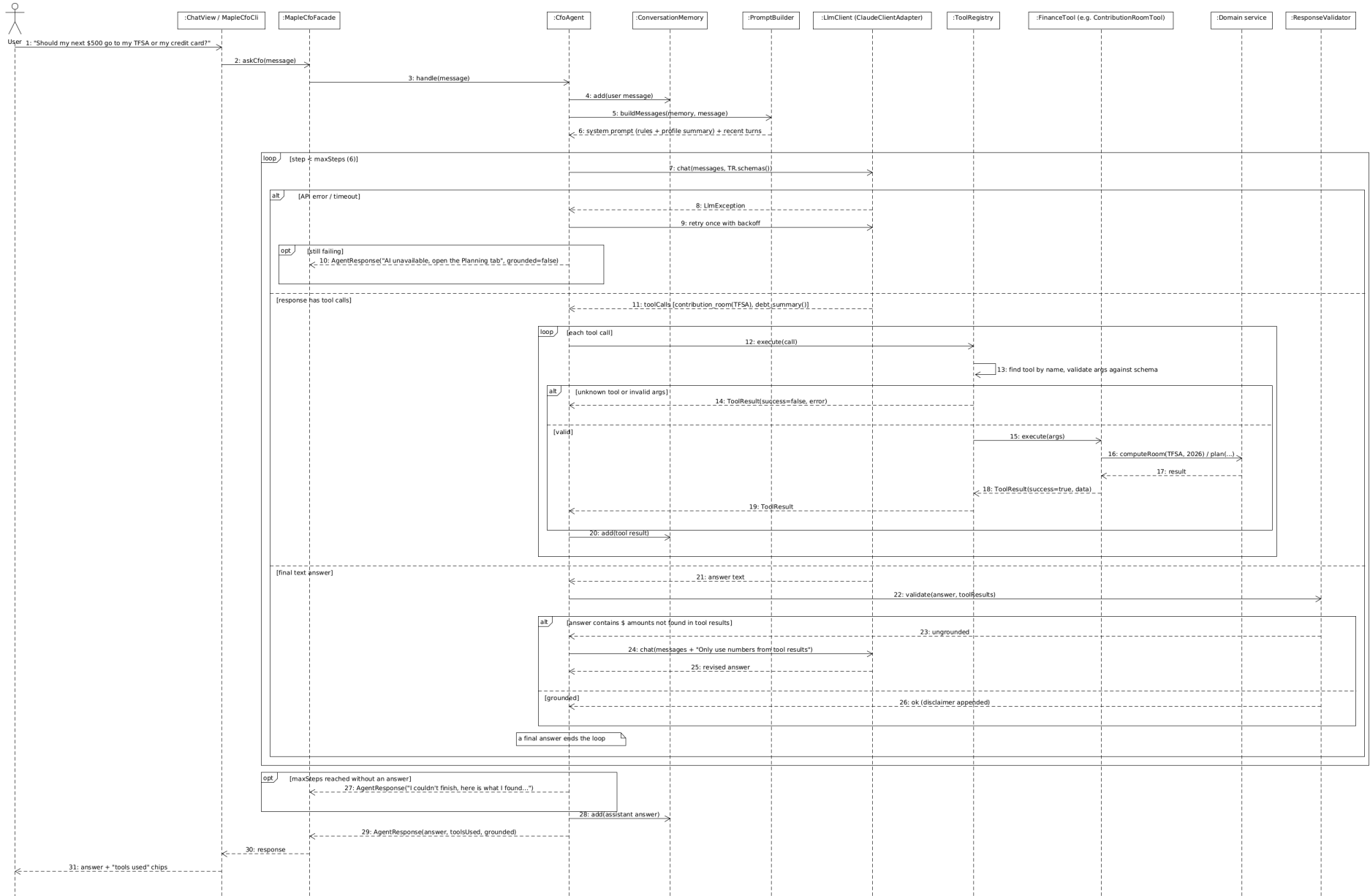
SD07: Create and forecast a savings goal (F09)



SD07: Create and forecast a savings goal (F09)

SD08: Ask the AI CFO (multi-step agent loop)

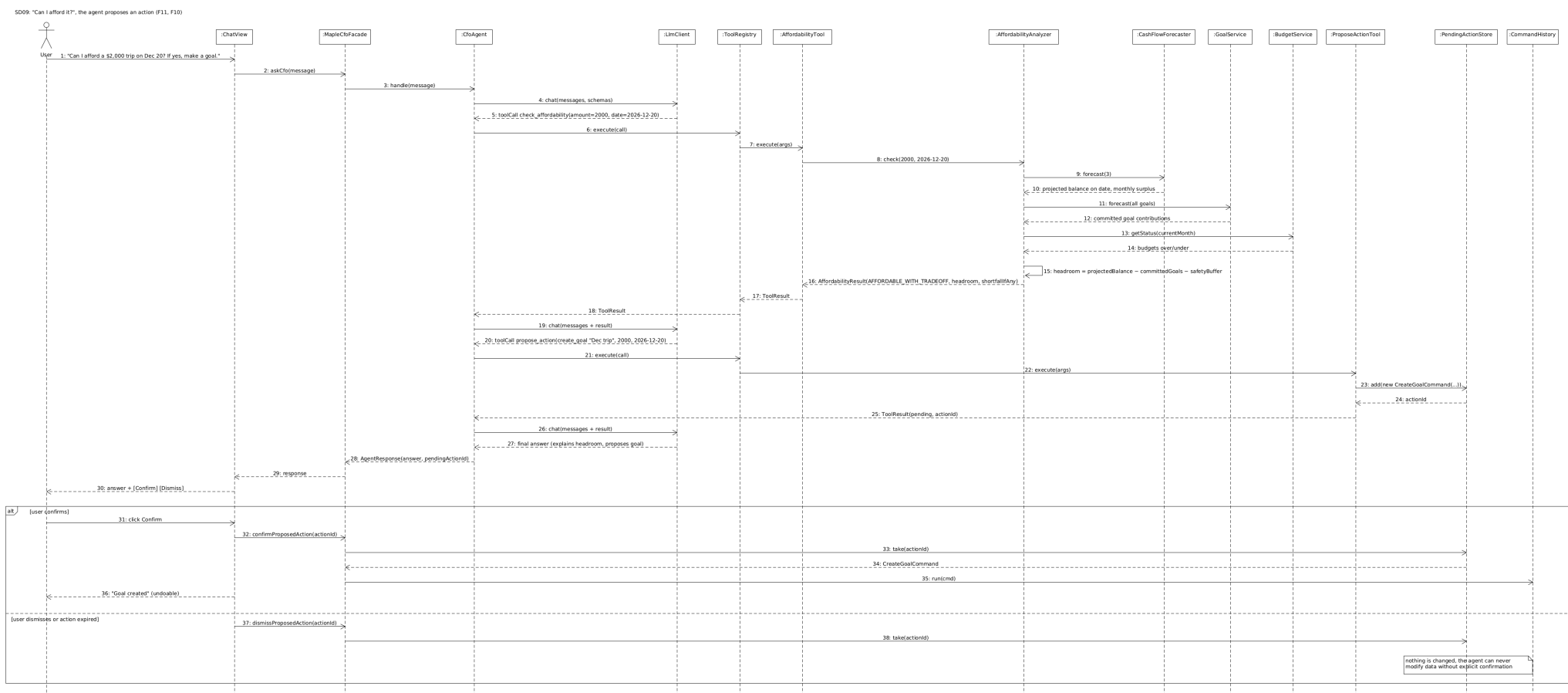
The core agent behaviour: reason → call tools → observe → answer, with argument validation, recovery from LLM and tool errors, a step limit, and the grounding check.



SD08: Ask the AI CFO, a multi-step tool-using agent loop (F10)

SD09: "Can I afford it?" with a proposed action

Planning across several steps and tools (affordability → propose a goal), and the **human-in-the-loop** confirmation implemented with Command objects.

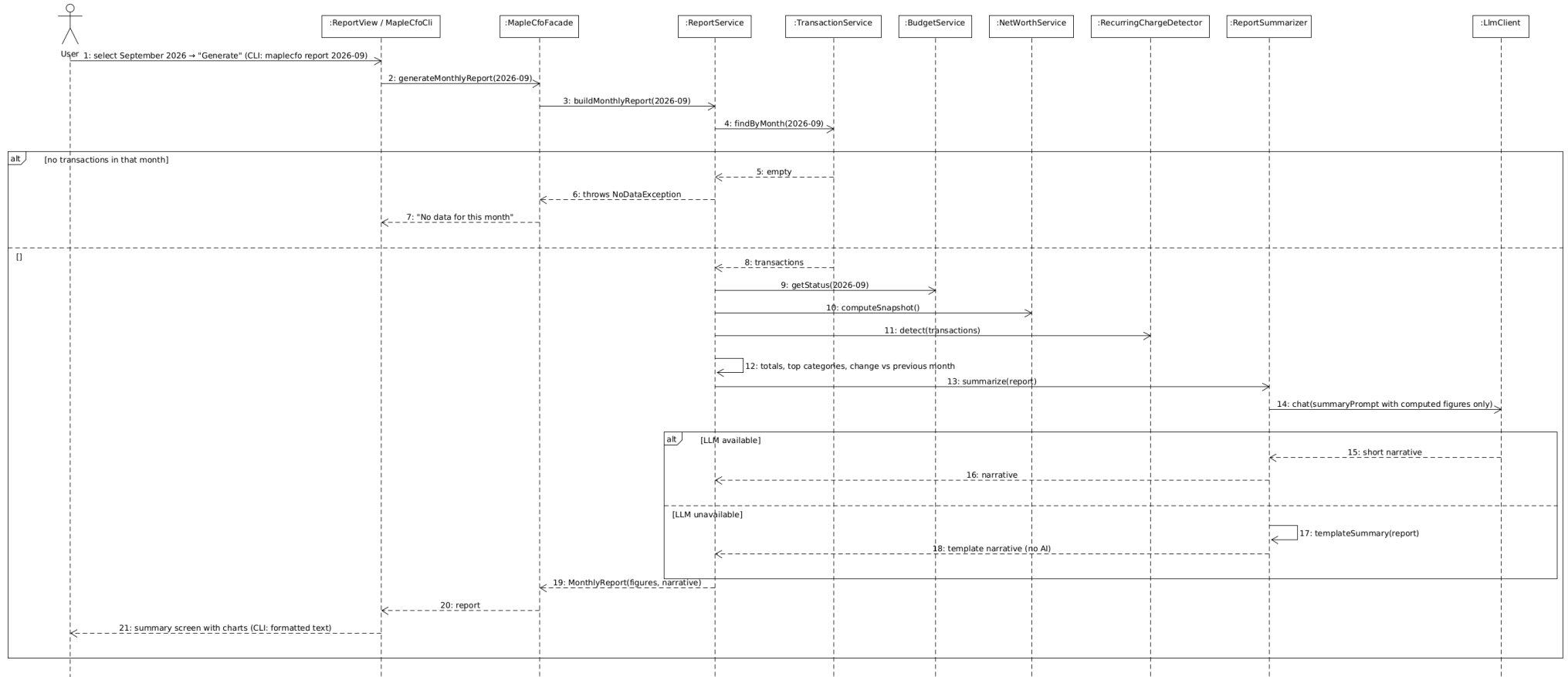


SD09: "Can I afford it?", the agent proposes an action (F11, F10)

SD10: Generate the monthly summary

The figures are deterministic, and the narrative comes from the LLM with a non-AI fallback.

SD10: Generate the monthly summary (F12)



SD10: Generate the monthly summary (F12)

8. Feature-to-design traceability table

Feature	Description	Type	Related use case	Classes	Key methods	Sequence diagram	Design pattern(s)
F01	Bank/card CSV import with mapping profiles	Deterministic	UC01	TransactionsView, MapleCfoFacade, TransactionService, MappingProfileRepository, ColumnMapping, TransactionImporter, CsvTransactionAdapter, TransactionRepository	importTransactions(), saveMappingProfile(), importFrom(), findByName(), isValidFor(), importFile(), existsByFingerprint()	SD01	Facade, Adapter, Repository
F02	Smart categorization with correction and undo	Hybrid	UC01, UC02	CategorizationService, RuleBasedCategorizer, LlmCategorizer, LlmClient, PromptBuilder, RecategorizeCommand, CommandHistory, TransactionService	categorizeAll(), categorize(), learnFromCorrection(), recategorize(), run(), undo(), redo()	SD01, SD02	Strategy, Adapter, Command
F03	Accounts and net-worth dashboard	Deterministic	UC03	DashboardView, MapleCfoFacade, AccountService, NetWorthService, AssetComponent, AccountGroup, Account, AccountRepository	updateAccountBalance(), buildAccountTree(), computeSnapshot(), getValue()	SD03	Composite, Facade, Repository
F04	Monthly budgets and alerts	Deterministic	UC04	BudgetView, BudgetService, SetBudgetCommand, CommandHistory, EventBus, AlertService, FinanceEventListener	setBudget(), getStatus(), onEvent(), publish(), subscribe()	SD01, SD02	Observer, Command
F05	Recurring charge detection	Deterministic	UC05	DashboardView, MapleCfoFacade, TransactionService, RecurringChargeDetector	getRecurringCharges(), findByMonth(), detect()	SD06	Facade
F06	TFSA / RRSP / FHSA room tracker	Deterministic	UC06	PlanningView, RegisteredAccountService, ContributionLimits, ProfileRepository, ContributionRepository, EventBus	checkContribution(), computeRoom(), tfsaLimitFor(), loadFrom()	SD04	Facade, Observer, Repository
F07	FIRE projection	Deterministic	UC07	PlanningView, MapleCfoFacade, FireCalculator	projectFire(), project(), yearsToFire()	SD05	Facade
F08	Debt payoff planner	Deterministic	UC08	PlanningView, DebtPayoffPlanner, PayoffStrategy, AvalancheStrategy, SnowballStrategy, DebtRepository	planDebtPayoff(), setStrategy(), plan(), orderDebts()	SD05	Strategy

Feature	Description	Type	Related use case	Classes	Key methods	Sequence diagram	Design pattern(s)
F09	Savings goals and forecasts	Deterministic	UC09	GoalsView, GoalService, CreateGoalCommand, CommandHistory, CashFlowForecaster	createGoal(), forecastGoal(), forecast(), recordDeposit()	SD07	Command, Observer
F10	AI CFO chat (agent)	AI	UC10	ChatView, MapleCfoCli, CfoAgent, ConversationMemory, PromptBuilder, LlmClient, ClaudeClientAdapter, ToolRegistry, FinanceTool, AbstractFinanceTool, ResponseValidator, PendingActionStore	askCfo(), handle(), buildMessages(), chat(), schemas(), execute(), doExecute(), validate()	SD08	Adapter, Template Method, Factory Method, Command, Facade
F11	"Can I afford it?"	Hybrid	UC11	GoalsView / ChatView, AffordabilityTool, AffordabilityAnalyzer, CashFlowForecaster, GoalService, BudgetService, ProposeActionTool, PendingActionStore, CommandHistory	checkAffordability(), check(), forecast(), confirmProposedAction(), take(), run()	SD09	Command, Template Method, Facade
F12	Monthly summary with AI narrative	Hybrid	UC12	ReportView, MapleCfoCli, ReportService, ReportSummarizer, LlmClient, BudgetService, NetWorthService, RecurringChargeDetector	generateMonthlyReport(), buildMonthlyReport(), summarize(), templateSummary()	SD10	Facade, Adapter

9. Feature implementation explanations

F01: Bank/card CSV import with column-mapping profiles

Related use case: UC01 · **Related sequence diagram:** SD01

Classes involved

- `TransactionsView`: collects the file, account and profile, shows the mapping dialog when needed, and shows the `ImportResult`.
- `MapleCfoFacade`: the single entry point, `importTransactions()` and `saveMappingProfile()`.
- `TransactionService`: coordinates profile lookup, import, categorisation, saving and events.
- `MappingProfileRepository` and `ColumnMapping`: saved column mappings, one per bank layout.
- `CsvTransactionAdapter`: adapts the CSV layout to `Transaction` objects (read, parse, validate).
- `TransactionRepository`: stores transactions and checks fingerprints.

Important methods: `TransactionsView.onImportClicked()`, `MapleCfoFacade.importTransactions()`, `TransactionService.importFrom()`, `MappingProfileRepository.findByName()`, `ColumnMapping.isValidFor()`, `CsvTransactionAdapter.importFile()`, `TransactionRepository.existsByFingerprint()`.

Execution: When the user clicks Import, the view calls `importTransactions()` on the facade, which delegates to `TransactionService.importFrom()`. The service loads the named `ColumnMapping` and checks it against the file's header with `isValidFor()`. If there is no profile, or it doesn't match, the view opens the mapping dialog and the new mapping is saved with `saveMappingProfile()`. The service then creates a `CsvTransactionAdapter` with the mapping and calls `importFile()`, which reads, parses and validates the rows. The resulting transactions are categorised (F02), duplicates are skipped using fingerprints, the rest are saved, and `TRANSACTIONS_IMPORTED` is published. The `ImportResult` goes back through the facade to the view.

F02: Smart categorization with correction and undo

Related use cases: UC01, UC02 · **Related sequence diagrams:** SD01, SD02

Classes involved

- `CategorizationService`: runs an ordered list of strategies.
- `RuleBasedCategorizer`: merchant and keyword rules, including rules learned from corrections.
- `LlmCategorizer`: asks the LLM for a category name and rejects anything that is not a valid `Category`.
- `PromptBuilder`: builds the categorisation prompt.
- `LlmClient`: sends the request.
- `RecategorizeCommand`, `CommandHistory`: an undoable user correction.
- `TransactionService`: applies the change and publishes the event.

Important methods: `CategorizationService.categorizeAll()`, `CategorizationStrategy.categorize()`, `RuleBasedCategorizer.learnFromCorrection()`, `MapleCfoFacade.recategorize()`, `CommandHistory.run()` / `undo()` / `redo()`, `TransactionService.recategorize()`.

Execution: During import, `categorizeAll()` passes each transaction to `RuleBasedCategorizer` first. If no rule matches, `LlmCategorizer` builds a prompt and calls `LlmClient.chat()`, then `parseCategory()` accepts the reply only if it is exactly one of the `Category` names; otherwise the transaction is flagged for review. When the user corrects a row, the facade creates a `RecategorizeCommand` and gives it to `CommandHistory.run()`. The command calls `TransactionService.recategorize()` (which returns the old category, kept for undo) and `CategorizationService.learn()` adds a rule. Undo calls the command's `undo()`, which restores the old category.

F03: Accounts and net-worth dashboard

Related use case: UC03 · **Related sequence diagram:** SD03

Classes involved: `DashboardView` (shows the net worth), `AccountService` (account CRUD, builds the Composite tree), `NetWorthService` (computes the snapshot), `AssetComponent` / `AccountGroup` / `Account` (Composite), `AccountRepository` (storage).

Important methods: `MapleCfoFacade.updateAccountBalance()`, `AccountService.buildAccountTree()`, `NetWorthService.computeSnapshot()`, `AssetComponent.getValue()`.

Execution: An edited balance is validated and saved by `AccountService`. `getNetWorth()` then calls `NetWorthService.computeSnapshot()`, which gets the root `AccountGroup` from `buildAccountTree()` and calls `getValue()`. Each group recursively adds up its children, and each account returns its balance (negative for liabilities). The snapshot, with the total and the per-group subtotals, is displayed and stored for the history chart.

F04: Monthly budgets and overspending alerts

Related use case: UC04 · **Related sequence diagrams:** SD01 (alerts), SD02 (undoable set)

Classes involved: `BudgetView`, `SetBudgetCommand`, `CommandHistory`, `BudgetService` (status computation, and an observer of transaction events), `EventBus`, `AlertService` and the views (observers of `BUDGET_EXCEEDED`).

Important methods: `MapleCfoFacade.setBudget()`, `BudgetService.setBudget()` / `getStatus()` / `onEvent()`, `EventBus.subscribe()` / `publish()`.

Execution: Setting a limit creates a `SetBudgetCommand`, which saves the budget and remembers the previous one for undo. At start-up `BudgetService` subscribes to transaction events. When `TransactionService` publishes `TRANSACTIONS_IMPORTED` or `TRANSACTION_RECATEGORIZED`, `BudgetService.onEvent()` recomputes `getStatus()` for the month. For any category at or above its threshold it publishes `BUDGET_EXCEEDED`, which `AlertService` and `BudgetView` / `DashboardView` receive to show banners, without `TransactionService` knowing about any of them.

F05: Recurring charge detection

Related use case: UC05 · **Related sequence diagram:** SD06

Classes involved: `DashboardView`, `MapleCfoFacade`, `TransactionService` (history), `RecurringChargeDetector` (the algorithm).

Important methods: `MapleCfoFacade.getRecurringCharges()`, `TransactionService.findByMonth()`, `RecurringChargeDetector.detect()`.

Execution: The facade loads the last six months of transactions and passes them to `detect()`. The detector normalises merchant names, groups the transactions, keeps groups with at least three charges at a regular interval, computes the average amount and next expected date, and flags price increases above 5%. The `RecurringCharge` list is shown in the Subscriptions panel. It is also used by `CashFlowForecaster` (F09, F11) and by the monthly report (F12).

F06: TFSA / RRSP / FHSA contribution-room tracker

Related use case: UC06 · **Related sequence diagram:** SD04

Classes involved: `PlanningView`, `RegisteredAccountService` (rules), `ContributionLimits` (yearly limits loaded from JSON), `ProfileRepository`, `ContributionRepository`, `EventBus`.

Important methods: `MapleCfoFacade.checkContribution()` / `getContributionRoom()`, `RegisteredAccountService.computeRoom()` / `checkContribution()` / `recordContribution()`, `ContributionLimits.loadFrom()` / `tfsaLimitFor()`.

Execution: `checkContribution()` loads the `UserProfile`. If it is incomplete, it returns `UNKNOWN` with the missing fields. Otherwise `computeRoom()` adds up the annual limits from `ContributionLimits` for every eligible year, adds withdrawals from previous years and subtracts contributions from `ContributionRepository`. The amount is compared with the room: `OK` returns the room before and after, while `OVER` returns the excess and publishes `CONTRIBUTION_WARNING`. Because the limits come from a file, updating them each year needs no code change.

F07: FIRE projection

Related use case: UC07 · **Related sequence diagram:** SD05

Classes involved: `PlanningView`, `MapleCfoFacade`, `FireCalculator` (pure function).

Important methods: `MapleCfoFacade.projectFire()`, `FireCalculator.project()` / `yearsToFire()`.

Execution: The view builds a `FireInputs` record and calls `projectFire()`. `FireCalculator.project()` validates the ranges, computes the target (spending ÷ withdrawal rate) and simulates year by year until the balance reaches the target or 70 years pass. It repeats the simulation for the scenario savings rates and returns a `FireProjection`, which the view charts. The calculator has no dependencies, so it can be tested with exact expected values.

F08: Debt payoff planner

Related use case: UC08 · **Related sequence diagram:** SD05

Classes involved: `PlanningView`, `MapleCfoFacade`, `DebtPayoffPlanner` (context), `PayoffStrategy` with `AvalancheStrategy` and `SnowballStrategy`, `DebtRepository`.

Important methods: `MapleCfoFacade.planDebtPayoff()`, `DebtPayoffPlanner.setStrategy()` / `plan()`, `PayoffStrategy.orderDebts()`.

Execution: The facade maps the chosen `PayoffMethod` to a strategy object and calls `setStrategy()`, then `plan(extra)`. The planner loads the debts, asks the strategy for the payoff order, and simulates month by month (interest, minimum payments, extra payment to the first debt in the order, and freed payments rolled into the next one). It returns a `PayoffPlan`. The view calls the planner a second time with the other strategy to show the comparison.

F09: Savings goals with forecasts

Related use case: UC09 · **Related sequence diagram:** SD07

Classes involved: `GoalsView`, `MapleCfoFacade`, `CreateGoalCommand`, `CommandHistory`, `GoalService`, `CashFlowForecaster`, `EventBus`.

Important methods: `MapleCfoFacade.createGoal()` / `forecastGoal()`, `GoalService.createGoal()` / `forecast()` / `recordDeposit()`, `CashFlowForecaster.forecast()`.

Execution: Creating a goal runs a `CreateGoalCommand` through `CommandHistory`, so it is undoable. `GoalService` validates and saves the goal and publishes `GOAL_UPDATED`. `forecast()` asks `CashFlowForecaster` for the average monthly surplus, computes the monthly amount needed to hit the target by the deadline, and returns a `GoalForecast` (on track or not), which the view shows as a progress ring.

F10: AI CFO chat (multi-step, tool-using agent with memory)

Related use case: UC10 · **Related sequence diagram:** SD08

Classes involved

- `ChatView` / `MapleCfoCli`: the interface.
- `MapleCfoFacade.askCfo()`: the entry point.
- `CfoAgent`: the agent loop.

- `ConversationMemory`: recent turns and tool results.
- `PromptBuilder`: system rules and profile summary.
- `LlmClient` (e.g. `ClaudeClientAdapter`, created by `LlmClientFactory`): the model call.
- `ToolRegistry`: finds tools, validates arguments, returns their schemas.
- `AbstractFinanceTool` and its 10 subclasses: wrap the deterministic services.
- `ResponseValidator`: checks the answer is grounded and adds a disclaimer.
- `PendingActionStore`: holds proposed commands.

Important methods: `CfoAgent.handle()` / `runLoop()`, `PromptBuilder.buildMessages()`, `LlmClient.chat()`, `ToolRegistry.schemas()` / `execute()`, `AbstractFinanceTool.execute()` / `doExecute()`, `ResponseValidator.validate()`, `ConversationMemory.add()`.

Execution: `handle()` stores the user message and builds the messages. In a loop of at most six steps, it calls `LlmClient.chat()` with the tool schemas. If the response contains tool calls, each one goes through `ToolRegistry.execute()`, which finds the tool and runs its template `execute()`. That validates the arguments, calls `doExecute()` (which calls a service such as `RegisteredAccountService.computeRoom()`) and wraps the outcome in a `ToolResult`, including errors, so the LLM can recover. The results are added to memory and the loop continues. When the LLM returns text, `ResponseValidator.validate()` checks that every dollar amount appears in this turn's tool results. If not, it requests one revision. It then appends an educational disclaimer. The `AgentResponse` is returned through the facade. The LLM never computes figures itself and can change data only by proposing a command (see F11).

F11: "Can I afford it?" affordability check

Related use cases: UC11, UC10 · **Related sequence diagram:** SD09

Classes involved: `GoalsView` / `ChatView`, `AffordabilityTool`, `AffordabilityAnalyzer` (verdict), `CashFlowForecaster`, `GoalService`, `BudgetService`, `ProposeActionTool`, `PendingActionStore`, `CommandHistory`.

Important methods: `MapleCfoFacade.checkAffordability()` / `confirmProposedAction()`, `AffordabilityAnalyzer.check()`, `CashFlowForecaster.forecast()`, `PendingActionStore.add()` / `take()`, `CommandHistory.run()`.

Execution: From the form, the facade calls `AffordabilityAnalyzer.check()` directly. From chat, the LLM calls `check_affordability`, which `AffordabilityTool` passes to the same analyzer. The analyzer projects the balance up to the date, subtracts committed goal contributions and a safety buffer, and classifies the result. The LLM explains it and may call `propose_action`. `ProposeActionTool` then creates a `CreateGoalCommand` and stores it in `PendingActionStore`, returning an id. The view shows Confirm/Dismiss. Only `confirmProposedAction(id)` takes the command and runs it through `CommandHistory`, so it is user-approved and undoable.

F12: Monthly summary with AI narrative

Related use case: UC12 · **Related sequence diagram:** SD10

Classes involved: `ReportView` / `MapleCfoCli`, `ReportService`, `TransactionService`, `BudgetService`, `NetWorthService`, `RecurringChargeDetector`, `ReportSummarizer`, `LlmClient`.

Important methods: `MapleCfoFacade.generateMonthlyReport()`, `ReportService.buildMonthlyReport()`, `ReportSummarizer.summarize()` / `templateSummary()`.

Execution: `buildMonthlyReport()` loads the month's transactions and collects the deterministic figures from the services. If the month has no data, it raises `NoDataException` and the view shows a message. `ReportSummarizer.summarize()` sends only these figures to the LLM to write a short narrative, and falls back to `templateSummary()` if the LLM fails. The `MonthlyReport` is returned through the facade and displayed with charts in the GUI, or as formatted text in the CLI.

10. Important design decisions

#	Decision	Rationale	Alternative rejected
D1	The LLM never does arithmetic. Every number comes from deterministic services, exposed as tools.	Financial answers must be correct and reproducible. Deterministic code can be unit-tested (Stage 3, Part A). The LLM's role is planning and explanation (Part B).	Letting the LLM compute answers from raw transactions: cheaper to build, but it produces hallucinated or incorrect numbers.
D2	ResponseValidator grounding guard. Dollar amounts in the final answer must appear in this turn's tool results.	A deterministic safeguard against hallucinated figures, which is also a clear behavioural requirement to test with KUMA.	Trusting the prompt alone.
D3	Human-in-the-loop for any data change. The agent can only <i>propose</i> <code>UndoableCommand</code> s; the user confirms them.	Safety and trust: the AI can never silently change budgets or goals, and every confirmed action is undoable.	Letting the agent write directly through tools.
D4	GUI and CLI share <code>MapleCfoFacade</code>.	Meets the GUI + CLI requirement without duplicating logic, and guarantees identical behaviour in both.	Separate controllers per UI.
D5	Local-first data: CSV import + SQLite, no live bank connection.	Reliable, free, private, and easy to demo and test. Bank-aggregation APIs are paid, complex and region-limited.	Plaid or open-banking APIs.
D6	LLM behind an interface (<code>LlmClientAdapter</code>), with only two implementations: Claude and <code>MockLlmClient</code>.	The agent code is independent of Claude's JSON format. The mock makes agent-loop tests deterministic and free, and lets the app run in demo mode without an API key. Another provider can be added later with one new class.	Calling the Claude API directly from <code>CfoAgent</code> , or supporting several providers from day one.
D7	Rules before AI for categorisation; the AI must answer with a valid category name.	Cheaper, faster and predictable for common merchants. The LLM only handles the long tail, and invalid answers are flagged for review instead of guessed.	LLM-only categorisation.
D8	Contribution limits in a config file (<code>contribution_limits.json</code>).	CRA limits change yearly. Data, not code, should change.	Hard-coded constants.
D9	Bounded agent loop (max 6 steps) + one retry on LLM errors + schema validation of tool arguments.	Prevents runaway loops and costs, and handles malformed LLM output safely. These behaviours will be targeted by KUMA tests (tool failures, invalid arguments, step limits).	An unbounded ReAct loop.
D10	Education-only scope. The agent refuses specific investment or stock picks and adds a disclaimer.	Responsible AI in a financial domain. MapleCFO is not a licensed financial advisor.	Unrestricted advice.
D11	Money value object using <code>BigDecimal</code>.	Avoids floating-point rounding errors in totals.	<code>double</code> amounts.
D12	Lean scope: one generic CSV importer with saved mapping profiles, 12 features, on-screen summaries (no file export).	Keeps the project buildable and fully testable in the course timeline, while every required element (GUI, CLI, 10+ features, 5+ patterns, real agent behaviour) is still met.	Separate adapters per bank, a card-rewards optimizer and PDF/CSV export (possible future extensions).

10.1 How the design prepares for Stage 3 testing

- **Deterministic components (JUnit 5):** `FireCalculator`, `DebtPayoffPlanner` + strategies, `RegisteredAccountService`, `RecurringChargeDetector`, `BudgetService`, `AffordabilityAnalyzer`, `CsvTransactionAdapter` + `ColumnMapping`, `CommandHistory`, `EventBus`, `ToolRegistry` argument validation, `ResponseValidator`. Integration tests go through `MapleCfoFacade` with in-memory repositories and `MockLlmClient`.
- **Agent behaviour (KUMA):** candidate behavioural requirements include correct tool selection (e.g. contribution-room questions must call `contribution_room`), no invented figures (grounding), respecting user constraints ("under \$300"), never changing data without confirmation, recovering from tool errors, refusing out-of-scope investment picks, and asking for clarification on ambiguous requests.

10.2 Technology stack (planned)

Concern	Choice
Language	Java 21 (Maven build)
GUI	JavaFX 21
CLI	picocli
Persistence	SQLite via JDBC (<code>sqlite-jdbc</code>)
JSON	Jackson
LLM	Anthropic Claude Messages API over <code>java.net.http.HttpClient</code> ; <code>MockLlmClient</code> for tests and demo mode
Testing	JUnit 5, Mockito; KUMA for agent behaviour
Version control	Public GitHub repository (the same repository for all three stages)

10.3 AI assistance disclosure (Stage 1)

This design was developed with the help of an AI assistant (Anthropic Claude) for brainstorming project ideas, drafting feature specifications and producing the UML diagrams as UMLet (`.uxf`) files. The author reviewed and refined the scope, features, class responsibilities and pattern choices. AI use during implementation will be documented in the Stage 2 collaboration log, as the course requires.